RINEYA RESEARCH

Rineya × MiMo

Post-training research, evidence audit and experiment roadmap

Decision edition · 27 September 2026

Prove a useful learning effect before scaling the recipe.

A staged program for compact Rineya models: verified supervision, fixed-scope adaptation, independent preservation, and only then reinforcement learning or modular composition.

32 methods · 12 candidate pipelines · 35 observed branch refs

Primary planning envelope: approximately 344M parameters and 8 GiB VRAM. The 596M soup lineage is treated separately.

Research only. No native training, environment execution, model loading, repository modification or push was performed. Unmeasured quantities and unresolved identities remain explicit.

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The complete method assessments are in Appendix B. Source evidence and proposals are distinguished throughout.

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If I were running the Rineya project next

Executive decision

The most valuable next investment is to establish a small, verified learning effect on one exactly bound Rineya checkpoint—not to activate all 32 methods or ingest the entire MiMo release. Start with source/runtime qualification, a deterministic answer-verifier path, and a fixed-scope LoRA SFT pilot. Then isolate curriculum, preservation, and RL effects in that order. Architecture changes and inference optimizations belong on separate tracks until they solve a measured bottleneck.

The premise of the dataset strategy needs correction. The inspected public release is an **agentic task/environment release**, not an established corpus of millions of successful teacher reasoning traces. Its displayed total is **7,780 rows**. The card organizes tasks into software engineering, vulnerability reproduction, knowledge work, web development and symbolic music. Successful trajectories must be obtained and verified separately before they become distillation targets or preference pairs. [M1]

The existing repository is better prepared for controlled experiments than for a large native run. PRLC-001 already has a staged controller, real PyTorch GRPO/SFT update machinery, preservation replay, immutable checkpoint receipts and failure handling. Its validation record reports 90 scoped CPU tests and explicitly reports zero Rineya loads, zero GPU training runs and zero MiMo environment executions. This is useful infrastructure, not measured model improvement. [R1–R3]

The closest high-value method combination is therefore **verified short-target data + fixed rank-8 LoRA + independent retention replay + frozen evaluation**. Curriculum and MaxRL are plausible next factors, but neither should hide the absence of trainable signal. Per-skill adaptation and delta composition are a second, parallel research track after both donors independently acquire their intended skills. Low delta cosine cannot substitute for that test.

Evidence boundary and reading guide

VERIFIED SOURCE means an inspected repository file or primary publication supports the statement. **RECORDED MEASUREMENT** means a source reports a measurement; it was not reproduced here. **INFERRED** means an engineering conclusion derived from that evidence. **PROPOSED** means a new experimental choice. **UNMEASURED** means no result was obtained.

The remote audit is targeted, not a line-by-line audit of every branch. Thirty-five current branch heads were enumerated; selected controller, objective, qualification and composition documents were read at immutable commits. Branch commit dates, all historical branches, all native artifacts and all CI runs were not exhaustively inspected. The data inspection establishes release metadata and a visible schema example, not corpus-wide token distributions, correctness rates or decontamination. Attempts to retrieve raw data and several subset viewers did not yield processable data. Consequently, no token-count or reasoning-length estimate is presented as measured.

The method register preserves all 32 user terms. GPO remains unresolved: the intended primary paper could not be identified reliably. Deja Vu means contextual sparsity; Nexus means the common-minima optimizer; SWA bounded Replay means the project's sliding-window cache-reconstruction idea. CLARE is explicitly disambiguated from a robotics namesake.

1. Current repository capability map

The connected repository is `LUNCY579/ChatBubble`, private, with default branch `main` at `b0ca54bdc1b69e2b19d7baeca2e3fdde3ff0a7b2` when inspected. A current branch name proves that a ref exists; it does not prove the advertised mechanism works. Selected evidence is pinned in Appendix A and the reference register. [R0]

Capability	Classification	Evidence and consequence
Staged data eligibility, proportions, promotion and rollback	IMPLEMENTED + SYNTHETIC-ONLY	PRLC controller; reuse rather than rebuild. Native performance is unqualified. [R1–R2]
PyTorch policy-gradient/SFT mechanics	IMPLEMENTED + SYNTHETIC-ONLY	Real tiny-model autograd tests and inspected objective source; not pretrained gradient qualification. [R2–R3]
Local standard-HF LoRA loader	IMPLEMENTED + SYNTHETIC-ONLY	Safetensors, local-only, no remote code; custom packed Rineya formats are not automatically supported. [R1]
Native selected-token likelihood/EOS parity	BLOCKED	Required by both PRLC and Q0; exact checkpoint/runtime receipts absent in inspected evidence. [R1,R7]
Independent preservation CE plus reference KL	IMPLEMENTED + SYNTHETIC-ONLY	Explicit rehearsal loss is distinct from all-correct RL replay prompts. [R1]
Durable optimizer/RNG/checkpoint resume	IMPLEMENTED + SYNTHETIC-ONLY	PRLC reservations, hash-linked receipts, atomic latest pointer and failure handling. Native interruption behavior unmeasured. [R1]
Automatic layer/rank unlocking	PLANNED / NOT ENABLED	PRLC freezes rank-8 q/v scope for a run and rejects excessive generic trainable scope. [R1]
MiMo task import and quarantine	PARTIALLY IMPLEMENTED	Specifications/revision preserved, but required domain capabilities remain unavailable. [R1]
Live MiMo multi-turn agents and graders	MISSING in inspected runnable path	No qualified Docker, visual, music or cyber execution path. Generic backend interfaces are not implementations. [R1]
MAXRL-COMP-002 source/data qualification	IMPLEMENTED; NATIVE BLOCKED	Integrity and source tests, not native generation or a training launcher. [R7]
Soup and delta-composition mechanics	IMPLEMENTED + SYNTHETIC-ONLY	SOUP-MINI/CSS finite-tensor mechanics; native collectors remain closed. [R5–R6]
RoLA × NGU integration	IMPLEMENTED + SYNTHETIC-ONLY	Logical-async random CPU fixture, not concurrent native throughput or pretrained efficacy. [R8]
Fresh final capability/retention certification	BLOCKED	New task banks, identities, margins and authorized native measurements still required. [R1,R5–R8]

No inspected new post-training component qualifies for **IMPLEMENTED + NATIVE-VALIDATED** for the proposed MiMo program. That does not imply the entire historical project lacks native evidence. It means historical scores and adjacent loader qualifications cannot be silently transferred to this run.

1.1 Reuse the architecture, not every default

PRLC separates task schemas and data (`schema.py`, `data.py`), curriculum and transitions (`curriculum.py`, `engine.py`), policy/update backends (`objectives.py`, `backends.py`), and evidence/accounting (`evidence.py`, `cli.py`). Its source selection, difficulty bands and retention fraction are different controls. The unlocked fraction counts compatible training **parent groups within a source**, not a percentage of all raw rows. That distinction is critical when different tasks share variants or environments. [R1]

Its exact verifier accepts a completed, EOS-terminated, stripped exact answer. It does not accept arbitrary free-form CoT and silently extract a final number. Therefore a short-rationale experiment needs a separately specified and qualified response envelope/verifier, or a different audited SFT path. A report configuration is not a ready-to-run existing CLI profile.

PRLC's actual policy primitive computes population-standardized group advantages, sequence-mean clipped token-ratio loss and a sampled frozen-reference KL term. Its Q0 sibling documents an inherited estimator using sample standard deviation for GRPO and a mean-normalized MaxRL convention. Mixing helpers would change the objective even if both are labeled GRPO. Freeze a single reduction, normalization, mask and clipping contract before comparing algorithms. [R3,R7]

1.2 Merge/reuse decisions

Use PRLC as the candidate control-plane foundation. Reuse Q0's exact source binding, causal-shift, padding/EOS and parameter-inventory checks by an explicit adapter—not by dropping files into a source-hashed package. Q0 specifically warns that adding Python files to its audited sibling changes the source inventory. Reuse SOUP-MINI's compatibility/ancestry concepts for common-base donor experiments, and CSS's frozen selection/evaluation concepts for composition. [R1,R5–R7]

Do not wholesale merge RoLA, NGU, MaxRL, recurrence and CSS into the first training branch. Their model scopes, numerical conventions and permissions differ. Keep separate experiment registries and opt-in adapters. Likewise, the two current titan-clare heads require a future content/diff review; their existence does not identify a canonical implementation.

1.3 What existing retention evidence actually says

SOUP-MINI's README reports an inherited attenuation trade-off: an unrounded Titan delta at 0.50 acquired 219/384 and retained 42/47, versus stored Titan at 234/384 and 39/47. The relevant lesson is a trade-off between new and retained behavior, not lossless acquisition; the small preservation denominator also limits precision. These numbers were not independently remeasured for this report. [R6]

The RoLA/NGU source review further records draw-dependent repair and a negative abstention confirmation. This undermines a blanket premise that small or sparse updates cannot forget. Its own protocol keeps pruning as a separate postprocessing factor after unpruned training arms. Adopt that separation, not the optimistic premise. [R8]

2. MiMo dataset anatomy and eligibility

2.1 Release identity and storage

The observed dataset-tree head is `639865fd3374018d6cb29b9fb82dd531406fcf5f`. Freeze the actual revision and recheck file hashes before enrollment; mutable `main` pages and cached viewer pages can reflect different moments. The file tree reports about **12.1 GB**, which is a file/storage measure, not a number of training tokens or all required Docker storage. [M2]

Root files include `code.parquet` (~13.3 MB), `cyber.parquet` (~228 kB), `music.parquet` (~135 kB), `webdev.parquet` (~4.12 MB), and an image mapping file. The `general` directory contains a roughly 3.12 MB training parquet and an `envs` directory. This structure already shows why dividing total gigabytes by a text-token conversion factor would be misleading. Sizes are displayed, rounded metadata, not downloaded-byte measurements. [M2–M4]

2.2 Real taxonomy and proposed eligibility

Release family	Observed evaluation style	Proposed first-use policy
Code / software engineering	Executable tests	Later, isolated sandbox and short patch tasks after tool-action qualification.
Cyber / vulnerability reproduction	Rule checks	Exclude from the first run; requires dedicated authorized sandbox and scope review.
General / knowledge work	Rubric judging	Use only after rubric, tool and factual-verification audit; do not assume deterministic reward.
Visual / web development	Visual grading	Not a text-only drop-in; image/renderer/judge costs are separate.
Music / symbolic composition	Rule checks	Specialized symbolic task, not a proxy for general reasoning.

The left two columns follow the official card; the eligibility decisions are this report's proposals. [M1]

The visible code schema contains `data_source`, `ability`, `agent_name`, a role/content `prompt`, `reward_model`, and `extra_info`. One shown code task has a user prompt, a rule-style reward descriptor with an empty ground-truth string, and serialized environment information such as working directory, Docker image and problem statement. That is not a chosen/rejected pair or a successful teacher rollout. The visible examples include more than one language, but no language percentages were measured. [M1]

2.3 Questions that remain unmeasured

Exact per-config row counts, parent-family counts, duplicate rates, tokenizer-specific prompt percentiles, total text tokens, task difficulty, reward reliability and successful-trajectory lengths remain unmeasured. The displayed train split is not an independently vetted evaluation split. A sample's empty ground-truth field must not be interpreted as a negative label; an external grader may supply the outcome.

Answer-length, reasoning-length, self-correction and teacher-attempt distributions cannot be inferred from a task specification alone. They become meaningful after collecting a trajectory dataset. Similarly, schema field names and a release card cannot establish the hidden selection process, teacher policy, acceptance

rate, or full MiMo training curriculum. The public release does not reveal everything about the stronger model's training.

The package includes a bounded offline JSONL/Parquet profiler. It reports the population it actually read, file hashes, field frequencies, domain counts, duplicate normalized prompts, character distributions and optional local-tokenizer prompt counts. It does not execute tasks, load a model, infer missing answers, download a tokenizer or declare semantic decontamination. Its included test is synthetic and is labeled as such.

2.4 License and provenance gate

The dataset card declares Apache-2.0. That label is not evidence that every bundled repository, dependency, image or generated output has identical downstream conditions. Before redistribution, inspect the release license and notices, environment/repository licenses and any teacher service terms used for newly generated trajectories. Preserve source revision, original task identity, derivative lineage and license notices in the training manifest. This is a provenance requirement, not a legal determination about every component. [M1]

2.5 The useful MiMo-to-student bridge

Treat the raw release as a **source of task structures and evaluators**. Build a separate trajectory store with immutable task/environment revision, policy identity, exact prompt tokens, each model action, tool observations, behavior probabilities where applicable, termination reason, verifier outputs and full resource cost. A successful trace may enter SFT only after its tool outputs and action suffixes are separated. A preference pair requires two defensibly ranked traces for the same task. An RL episode requires a live or reproducible environment and a qualified reward path.

For a tiny student, start with an original, short, deterministic bridge curriculum: simple symbolic transformations, bounded instruction constraints, short algorithmic traces and later small testable code changes. These are **not MiMo rows** unless they actually come from the release, and they are **not a reproduction of MiMo's hidden generation process**. Their purpose is to qualify the learning mechanism cheaply before testing transfer to full agent tasks.

3. What stronger-model behavior can transfer?

Distillation can transfer a response distribution; it does not automatically transfer the teacher's entire internal computation. A student can reproduce phrases such as verification or backtracking without learning when they are necessary. Long teacher reasoning also consumes training targets that may exceed the student's useful context or representational capacity. This is a risk to test, not a theorem that long reasoning never helps. Published distilled reasoning models motivate the experiment but do not establish the same result at 344M. [P31]

The research target should be explicit: improved correctness on new problems at an acceptable cost, with bounded loss of old capabilities. Measure answer-only accuracy, changed task wording, altered surface templates, source-held-out problems and perturbed distractors. Report output length and compute alongside correctness. A longer explanation is not itself evidence of acquisition; a lower training loss is not evidence of generalization.

3.1 Distillation choices

Verified sequence distillation is the default: train on successful, short-enough teacher or program-generated responses with provenance. It needs no teacher co-resident in GPU memory. A concise response and a longer rationale can be paired to study reasoning length, but simplification must be independently checked; deleting intermediate text is not a proof-preserving transformation.

Soft-target distillation uses teacher distributions rather than only emitted tokens. Forward-KL-style matching penalizes the student for ignoring teacher-supported alternatives; reverse-KL-style objectives can favor concentration on modes the student already models. The exact loss, sampling distribution and tokenizer alignment matter. A full dense logit cache can be enormous: at 150,000 vocabulary entries, 1 million positions and two bytes per logit, the raw array is about 300 GB. This is an illustrative arithmetic calculation, not a measured Rineya cache. Prefer sequence targets first; a sparse logit cache needs a mathematically specified treatment of omitted probability mass.

Hidden-state or tangent distillation needs aligned internal representations and a qualified forward/JVP path. CSS contains generic mechanics, not native Qwen3 qualification. Its use should answer a narrow question about composition, not be presumed necessary for ordinary SFT. [R5]

Multi-teacher collection can expand coverage, but teacher routing must use training-side signals. Count teacher calls and verifier costs, retain disagreements, and independently check a sample of accepted outputs. A teacher's confidence is not a correctness label.

3.2 Preference and RL eligibility

DPO is sensible only after obtaining defensible same-prompt comparisons. An environment specification, a reward-function name or an empty answer field is not preference supervision. First compare positive-only SFT with pairwise training on the same prompt population; otherwise an apparent DPO advantage may simply reflect more useful data. [P05]

RL requires useful exploration support. For independent binary-success draws with success probability p and group size G, the probability of at least one success is `1-(1-p)^G`; the probability of a mixed group is `1-p^G-(1-p)^G`. At p=0.01 and G=4, both are about 3.94%; most groups contain no positive learning contrast. Increasing G spends more generation but does not make impossible tasks teachable. These are mathematical diagnostics, not measured student success rates.

Use GRPO as the simplest inherited policy-gradient reference, then compare it with rejection-SFT and MaxRL under equal **total** generation budgets. Do not combine a new estimator, a new adaptive sampler and a new attention architecture in the first comparison. [P03,P30]

3.3 Additional methods worth keeping as controls

High-quality SFT and verified rejection sampling deserve explicit status as serious methods, not merely weak warmups. LIMA supports the importance of selective alignment data in its setting; it does not establish that a few thousand examples will solve a much smaller student's reasoning bottleneck. DeepSeek-R1 supplies a stronger-model distillation precedent, but the capability and capacity gap remains an experimental variable. [P31–P32]

QLoRA is a memory strategy to compare only when the measured BF16 LoRA baseline does not fit comfortably. It introduces quantized base representations and associated numerical/runtime choices. It is not evidence that lower precision improves acquisition. Selective-layer tuning is another legitimate capacity control: compare q/v adapters, a broader adapter scope and one explicitly selected transformer block under matched data and evaluation, while reporting their different optimizer and activation costs. [P29]

ScaleRL provides a useful reason to examine loss aggregation, normalization, filtering and off-policy details rather than treating all “RL” as one recipe. Its compute-scaling observations come from a very different budget; no fitted large-run asymptote is imported into a 344M pilot. Self-training work also shows why self-generated rewards require independent verification: apparent self-improvement can evolve into reward hacking. [P33–P34]

3.4 Mathematical distinctions to freeze in the implementation

For a verified response y, completion-only SFT minimizes minus the sum of log policy probabilities over its target tokens. State whether the outer average is over tokens or over sequences; long trajectories receive different influence under those choices. For paired preferences, DPO uses a logistic loss on the chosen-minus-rejected difference in policy/reference log-ratios. Neither operation can recover an absent successful target from a task descriptor. [P05]

For correctness probability $p_\theta(x)$, ordinary expected-reward training targets an average of $p_\theta(x)$, whereas the motivating maximum-likelihood RL objective averages $\log p_\theta(x)$. The latter changes prompt weighting, especially at low success probability. A finite sampled implementation needs its actual estimator and stopping rule; simply dividing rewards by an observed mean is not a proof of the full limiting objective. [P03]

A proposed preservation objective can combine acquisition loss, independent replay cross-entropy and an explicit reference divergence:

```
L = L_acquisition + lambda * L_replay_CE + beta * L_reference
```

This is a design family, not a statement that all three terms use interchangeable reductions or that a given coefficient has universal meaning. Replay can protect a labeled finite task population while KL anchors sampled behavior elsewhere. A gradient-projection constraint protects only directions and inputs represented by its constraints. Neither diagonal Fisher penalties, low-rank updates nor orthogonal parameter vectors imply semantic noninterference on unobserved tasks.

4. The method map

The 32 detailed assessments in Appendix B separate published mechanisms from proposed Rineya tests. Every record addresses assumptions, benefit, failure modes, forgetting and instability, compute/VRAM/RAM/storage, data, implementation, evaluation, interactions and falsification. The editable CSV preserves these fields for filtering.

Track	Methods	How to use them
First learning/control layer	LoRA, WSD, DPO, MaxRL, NGU, Reinforce-Ada, score centering, Nexus	Choose one objective and one sampling policy at a time; WSD is only a schedule.
Modularity and preservation	Delta weights, CLARE, JumpLoRA, F-WANDA	Establish independent donor competence, then compare functional trade-offs with matched dose/capacity.
New computation/architecture	Recurrence, RoLA, SoftToken, Looped Flow, DeepLoop, discrete diffusion, Full-Bandwidth Transformer, Explorative Modeling, EqR, MoBA, NSA	Separate conversion/stability programs; not mandatory ingredients in MiMo SFT.
Agent/data orchestration	ReAct, Combee, Dream-RSI	Can improve task execution or trace collection without changing model weights.
Inference systems	Deja Vu, SWA bounded Replay, PagedAttention, LoopSpec	Spend engineering effort only when a measured bottleneck and compatible target exist.
Identity unresolved	GPO for RL scaling	Recover exact paper/objective before allocating implementation or training.

An exhaustive Cartesian product is not a scientific plan. With merely 12 binary switches there are 4,096 combinations, many incompatible or impossible to attribute. Use dependency screening, single-factor tests, a small factorial for plausible interactions, and confirmation on a fresh evaluation bank.

5. Data selection and curriculum

5.1 Compare strategies without changing the question

Strategy	Decision for the first program	Decisive comparison
All usable MiMo	Do not start here; environment cost and heterogeneous supervision dominate.	Scale only after a demonstrably useful eligible subset exists.
Uniform filtered subset	Essential baseline.	Same verified examples, shuffled schedule.
Easy-to-hard curriculum	Plausible, not assumed better.	Same final examples/exposures and training-token budget; order changes only.
Skill-balanced subset	Good default against domain collapse.	Natural-frequency versus balanced sampling with fixed eligibility.
Intermediate student difficulty	Promising after reliable competence estimates.	Random versus difficulty weighting, including probe cost.
Teacher-correct/student-wrong	Useful acquisition targeting if teacher verified.	Matched source/length/diversity controls; not only highest loss.
Verified-only	Strong default for deterministic targets.	Audit false acceptance, not merely reward-field existence.
Small set repeated	Cheap pilot, but memorization risk grows.	New examples versus repeated exposures at equal tokens.
Progressively expanding access	Supported by PRLC mechanics, efficacy unproven.	Distinguish eligibility changes from order-only curriculum.
MiMo-derived traces + retention replay	Preferred eventual structure.	Replay and acquisition populations remain disjoint from scoring banks.

5.2 Practical selection algorithm — proposed

First define parent families and split them before scoring difficulty or generating variants. Use only TRAIN for student-selection measurements. Maintain per-task success statistics with a smoothing prior, sample count and policy-version timestamp. A simple Beta(1,1) posterior mean `(successes+1)/(attempts+2)` prevents treating 0/4 as an exact zero probability, although it does not make four samples highly informative. PRLC already uses this style of estimate. [R1]

Use three pools: well-mastered, intermediate and unresolved/hard. Reserve explicit exploration for unknown or stale tasks rather than permanently excluding them. A proposed training-side weighting is:

```
weight(i) = domain_quota(i) * diversity_weight(i)
            * [0.20 + 0.80 * 4 * p_hat(i) * (1-p_hat(i))]
```

This is a heuristic for sampling, not the MaxRL objective and not a proof of maximal learning progress. Keep a uniform exploration floor. Recompute only after a registered token interval, cap each parent family's share, and retain the realized sampling probabilities. A noisy grader can also produce p near 0.5, so verifier reliability precedes difficulty optimization.

For loss-based selection, compare response-normalized NLL within domain and length buckets. Raw total loss otherwise favors longer samples. High loss may mean a malformed template, wrong target, vocabulary mismatch or corruption—not useful difficulty. A model's correct-but-low-probability answer and a teacher's stylistically different answer also require different interpretation.

5.3 Reverse-engineering task-design principles

A defensible generator has separate task specification, instance sampler, solver/verifier, surface renderer, independent checker, lineage store and split allocator. Generate original latent instances from owned or permitted rules. Use teachers for explanations only after a solver determines an answer. Validate the final target independently, including negative controls and mutations that should fail.

Infer candidate pedagogical dimensions from the release—tool dependencies, number of constraints, environment state, required tests—not unobserved hidden teacher policies. Create a factorial task generator with controlled composition: one skill, another skill, both together, and novel surface forms. Hold out complete combinations or source families. This tests compositional transfer more directly than random string splits.

For real code environments, use isolated ephemeral sandboxes, disabled outbound network by default, resource/time limits, readonly protected files, no credentials, immutable images and logged tests. Do not execute raw dataset commands merely because they are present. Exclude cyber/visual/music tracks from the initial text bridge; each requires its own environment and grading qualification.

6. Twelve candidate pipelines

These are alternatives and staged research candidates, not a recommendation to run twelve campaigns now. Expected upside is a hypothesis, not a forecast of accuracy. Costs below use the sensitivity conventions in Section 9; no native training throughput was measured.

ID	Pipeline	Confidence / priority	Minimum informative pilot
C01	Verified short targets → fixed LoRA SFT → frozen check	Highest practical readiness / NOW	4,096 acquisition rows, 1,024 replay rows, one epoch; base versus trained.
C02	Same examples → easy-to-hard order → same LoRA	Plausible / NEXT	C01-sized paired random/curriculum arms, three seeds after a screen.
C03	Student competence probes → diversity-constrained selection → SFT	Plausible / NEXT	Score 8,192 training candidates; select 4,096; charge probe tokens.
C04	Verified successes → rejection-SFT with replay	Strong simple comparator / NEXT	512 prompts × 4 capped attempts; retain all attempt costs.
C05	SFT bootstrap → fixed-group GRPO + preservation CE	Plausible when reward signal exists / NEXT	512 verifiable prompts, G=4, ≤1M generated tokens/arm.
C06	Same bootstrap/tasks → MaxRL + preservation CE	Promising objective / NEXT	Matched C05 prompts and total sampling budget; exact estimator frozen.
C07	Same GRPO objective → bounded NGU sampling	Conditional hard-task value / NEXT	Cap attempts per parent and total tokens; compare fixed sampler.
C08	Same objective → Reinforce-Ada allocation	Conditional efficiency value / NEXT	Match C07 total tokens and task coverage; no uncounted discard.
C09	Verified same-prompt preferences → DPO → check	Conditional on pairs / NEXT	1,024 audited pairs plus matched positive-only SFT.
C10	Two skill-specific LoRAs → dose-controlled merge/routing → consolidation	High research relevance, uncertain advantage / NEXT	2,048 parents per skill; monolithic and unmerged endpoint controls.
C11	Same skill streams → JumpLoRA or CLARE-inspired isolation	Promising but added complexity / LATER	One isolation method versus dense LoRA; same order and budget.
C12	Bounded ReAct environments → verified teacher traces → compact bridge SFT	Necessary MiMo connection, high execution cost / LATER	32 sandbox-qualified code tasks, capped teacher attempts and tool steps.

C01–C03 are principally data/schedule comparisons. C04–C09 are objective or sampling comparisons only after source competence. C10–C11 ask whether modularity is worth its extra machinery. C12 establishes actual transfer toward the release's agent setting. None claims that teacher acquisition and verifier cost are free.

6.1 Monolithic versus modular training

Compare one mixed-data LoRA with two independently trained per-skill LoRAs from the same base. Equalize total **training** tokens across the program: two adapters each trained on 2,048 parents consume the same acquisition population as one adapter trained on all 4,096. Report the combined storage and inference routing cost. A routed ensemble receiving task labels or gold-derived oracle selection is not comparable to an answer-unaware deployed model.

The first composition screen needs base, each endpoint, each component at its intended dose, their sum/average and a norm-matched control. For same-base $W_i = W_0 + \Delta W_i$, averaging two endpoints gives $W_0 + 0.5\Delta W_1 + 0.5\Delta W_2$; a full delta sum uses coefficients 1 and 1 and changes the update magnitude. A benefit from the sum alone does not establish functional orthogonality. [R5–R6]

LoRA factors have a change-of-basis ambiguity: replacing B by BQ and A by $Q^{-1}A$ leaves BA unchanged. Therefore averaging A/B factors from independently trained adapters is not generally equivalent to averaging their effective updates. Materialize or combine the effective deltas using a qualified method.

Projection also needs a causal interpretation. For a fixed linear projection P, $P(a\Delta_1 + b\Delta_2) = aP\Delta_1 + bP\Delta_2$. If a compress-then-merge procedure beats merge-then-compress, the learned basis or rank allocation—not the algebraic ordering of a fixed projection—must explain the difference. CSS explicitly recognizes this. [R5]

7. Interaction matrix and forbidden shortcuts

All relationships below are engineering hypotheses unless a referenced implementation specifies the combination. A mechanically possible pair is not evidence of positive synergy.

Pair	Relationship	Reason and test
LoRA + verified SFT	Strongly complementary operationally	Modest optimizer state with direct supervision; capacity remains a hypothesis.
Curriculum + progressive unfreezing	Potentially complementary, initially confounded	Both change effective task difficulty; test separately, then a 2×2 factorial.
Retention CE + KL	Potentially complementary	CE preserves labeled tasks; KL anchors sampled behavior. Neither covers all inputs.
RL + all-correct replay prompts only	Insufficient preservation	Centered group advantage may be zero; keep independent CE. [R1]
MaxRL + NGU	Unknown / estimator-sensitive	Objective reweighting and adaptive sampling interact; derive or faithfully qualify the combination.
NGU + Reinforce-Ada	Potentially redundant/conflicting	Both allocate attempts; a combined rule may double-weight difficult tasks.
Score centering + importance correction	Potentially complementary	Addresses specific mismatch; cannot fix a wrong template or action distribution.
WSD + LoRA	Compatible	Schedule and parameterization differ; no reason to assume the best schedule transfers.
DPO + GRPO	Sequential alternative, not mandatory	Requires additional valid pairs and two objectives; compare simpler alternatives first.
CLARE + JumpLoRA	Mostly overlapping hypotheses	Both pursue sparse isolation; establish separate benefits before adding two mechanisms.
F-WANDA + delta composition	Unknown / worth a later test	Full-weight pruning evidence does not establish delta-semantic preservation.
Nexus + orthogonal skill deltas	Potentially conflicting assumptions	Encouraging gradient agreement differs from insisting update directions separate.
RoLA + dense attention	Adds work in the local residual design	Test quality; do not claim a sparse-attention speedup. [R8]
MoBA + NSA + RoLA	Alternative architecture designs	No justified default three-way attention replacement.
SoftToken + Looped Flow + EqR	Potentially redundant/complex	Three different latent-computation contracts; isolate one against an ordinary recurrent control.

Pair	Relationship	Reason and test
DeepLoop + recurrence	Dependency-aligned	Stabilization matters only under matching tied-depth/normalization assumptions.
LoopSpec + ordinary nonlooped Rineya	Incompatible as stated	Requires suitable early loop states and target verification.
PagedAttention + RL sampling	Operationally complementary	More efficient KV handling may fund rollouts; exact sampling/logprob contract must survive.
SWA replay + PagedAttention	Potentially complementary	Recomputation and allocation are different levers; combined speed/quality is unmeasured.
ReAct + MiMo	Task-interface alignment	Still requires sandbox, action schema, reward and multi-turn masks.
Combee/Dream-RSI + distillation	Upstream complement	Better trace collection is useful only if resulting traces are verified and transferable.

Do not interpret sparse storage as sparse GPU execution, a tiny update as harmless, low cosine as skill independence, an operational CAL interval as final significance, or source-tests passing as training permission.

8. Evaluation and ablation design

8.1 Splits and leakage controls

Create parent-family splits before rendering variants or generating teacher answers. Keep TRAIN, selection/training-side probes, DEV/CAL, CHECK and FINAL distinct. PRLC's calibration is repeatedly consulted for promotion and therefore becomes development data; it is not a final statistical holdout. Preserve historical banks as exposed/report-only unless their untouched provenance is actually established. [R1,R7]

Use exact and normalized prompt hashes, explicit parent IDs, source/repository lineage and near-duplicate review. MinHash or embeddings can flag candidates for inspection, but no similarity threshold proves the absence of contamination. In code, forks or variants of the same bug may share a parent even when prompts differ. A random re-split of an exposed bank does not restore independence.

8.2 Minimum evaluation panel — proposed

For the first short-task program, reserve at least 512 independent-parent CHECK acquisition items across the targeted skill families and a distinct 512-item preservation panel. Use existing qualified project banks only when their exposure and ancestry satisfy the plan; otherwise generate fresh original parents. Later confirmation should use 1,024 or more fresh acquisition and preservation items, with counts set by a power calculation using development discordances.

Measure paired pass@1 change; source-correct losses and source-incorrect repairs; per-family accuracy; completion/EOS failure; invalid answer format; output length; wall time; peak VRAM/RAM; total prompt, response, reference and gradient-processing tokens. Include simple instruction constraints, short symbolic reasoning and held-out compositions. For preservation, retain relevant retrieval, long-context, general-language and abstention panels separately; an easy arithmetic gain must not cancel a retrieval regression.

For confidence/calibration, prefer a defined answer-probability or classifier score with Brier/log-loss metrics. Do not call verbal confidence calibrated merely because the model emits percentages. Evaluate abstention both on truly unanswerable cases and false abstention on answerable cases. A formatting fix and a newly solved problem must be reported separately.

8.3 Statistical policy

The unit of independence is the parent task, not every surface variant, checkpoint or seed. Use paired parent-cluster resampling for descriptive intervals. For a confirmatory superiority claim, preregister the primary endpoint and multiplicity treatment. For preservation, choose an explicit noninferiority margin and show the corresponding lower bound; a nonsignificant difference is not proof of equivalence.

A useful planning approximation for paired binary outcomes is $SE \approx \sqrt{(q-d^2)/n}$, where q is the fraction of discordant pairs and d the accuracy difference. With q=0.10 and n=512, the approximate 95% half-width is 2.7 percentage points; at n=1,024 it is 1.9 points. These calculations illustrate why a 512-row pilot cannot reliably certify a 1-point preservation margin. They are not a power guarantee, and clustering can further reduce effective sample size.

Use one primary comparison per pilot, freeze the model-selection rule before CHECK, and open CHECK only after each arm's final checkpoint is committed. Multiple training seeds measure sensitivity to optimization; they do not magically multiply the number of independent evaluation tasks. Report per-seed effects rather than treating row×seed scores as independent rows.

8.4 Cheap-to-expensive tiers

Tier 0: data and identity. Schema/provenance census, deterministic verifier tests, split audit and exact checkpoint/tokenizer binding. Zero model training. Fail on ambiguous parents, unsupported modalities or answer leakage.

Tier 1: native mechanics. Frozen inference parity, raw sampled-token/logprob agreement, EOS/padding cases, scoped finite gradients and one explicitly authorized reversible optimizer/resume fixture. This is not an efficacy experiment. Keep test mutation outside production checkpoints.

Tier 2: acquisition pilot. C01 with a fixed final checkpoint, followed by confirmation seeds only if the initial operational screen is promising. Establish a learnable target before comparing sophisticated objectives.

Tier 3: factor isolation. C02 random order versus curriculum; then replay dosage or parameter scope. A later 2×2 curriculum/unfreezing design estimates interaction, but use an identical final token budget and model inventory rules.

Tier 4: objective/sampler comparison. GRPO versus rejection-SFT versus MaxRL; separately fixed sampling versus NGU or Reinforce-Ada. Match total generated work and include reference-scoring overhead.

Tier 5: confirmation and transfer. Fresh source families, multiple seeds, a powered preservation panel and a bounded MiMo agent transfer task. Scale only a frozen winner that survives both acquisition and preservation checks.

9. Hardware and cost model

9.1 What can be calculated without a benchmark

Use the reported 8 GiB envelope as a constraint, but record the actual GPU name and `total_memory` at qualification. Do not infer memory capacity from a marketing name alone. Model parameter count and tensor inventory must also be measured on the selected checkpoint, not copied from an adjacent family.

For N=344 million parameters, BF16 weights alone occupy approximately 0.688 GB (0.64 GiB). A conventional mixed-precision full-training layout with 2-byte weights, 2-byte gradients, 4-byte master weights and two 4-byte Adam moments is about 16N bytes: approximately 5.50 GB (5.13 GiB) before activations, logits, workspaces or allocator overhead. Layouts differ, so this is a component estimate, not a universal allocator measurement. At 596 million parameters the same layout is about 9.54 GB (8.88 GiB), already beyond an 8 GiB device before activations.

LoRA reduces trainable optimizer states but not the need to backpropagate through the network. Full-vocabulary logits, long contexts and KV caches can dominate. At vocabulary 150,000, length 1,024 and FP32 logits, one dense logits array is about 614 MB (0.57 GiB); gradients and intermediate buffers can add more. This is a sensitivity example, not an assertion of the exact selected model vocabulary.

KV tensor bytes are roughly `2 × layers × tokens × KV_heads × head_dim × bytes_per_element × batch`, excluding metadata and runtime buffers. Plug in the actual GQA inventory. A long group of eight concurrent trajectories can change the fit even though parameter count is constant. Serial rollout microbatching may be preferable locally.

9.2 Local versus larger-memory hardware

A short-context 344M LoRA pilot is the best local candidate; BF16, microbatch one, gradient accumulation and checkpointing should be qualified before adding quantization. QLoRA can reduce base storage, but it changes numerics and may add kernel overhead; for an already-small student it is a fallback to a measured memory problem, not an automatic first improvement. [P29]

A 24 GiB GPU provides room for longer activations, GPU-resident reference scoring or larger rollout concurrency. A 48 GiB device may simplify larger teachers or environment-heavy workflows. Neither size guarantees faster useful learning, and this report does not quote current rental prices or measured throughput. Host RAM may remain binding even after renting more VRAM if CPU references, images and environments run on the same constrained host.

Preserve the project's existing admission and worker RAM guards; do not lower them to make an experiment launch. Avoid loading all donor state dictionaries or all environment assets simultaneously. For soup construction, one FP32 artifact at a time is already the appropriate design. [R1,R6]

9.3 Throughput sensitivity — not a performance forecast

Let U be actual nonpadding tokens processed through training, Q the measured training-token throughput, and T_other the separately measured generation, reference, evaluation, verification and checkpoint time. Then `T_total ≈ U/Q + T_other`. For RL, track all of these terms directly; generation-only token rates are not training throughput.

Workload	Assumed rate	Computed kernel/work time, before other costs
1.31M training tokens	250 / 500 / 1,000 tokens/s	1.46 / 0.73 / 0.36 hours

Workload	Assumed rate	Computed kernel/work time, before other costs
3.93M training tokens (three seeds at that example size)	250 / 500 / 1,000 tokens/s	4.37 / 2.18 / 1.09 hours
1M generated rollout tokens	25 / 50 / 100 tokens/s	11.11 / 5.56 / 2.78 hours
512 prompts × 4 attempts × 256-token cap	Up to 524,288 generated tokens	5.83 / 2.91 / 1.46 hours at the same decode rates

These rates are deliberately hypothetical sensitivity points. They are not observed on the user's GPU and not comparative benchmarks for 8/24/48 GiB machines. The 1.31M example assumes 5,120 training sequences averaging 256 total tokens; the first pilot's actual mean may differ substantially. Report the exact token manifest before authorization.

For a token ceiling of 5,242,880 tokens per seed, corresponding to 5,120 sequences at a maximum 1,024-token context, pure training time at the same three rates is about 5.83, 2.91 or 1.46 hours. Evaluation, CPU reference scoring and wall-clock stalls remain additional. A “small number of rows” is not a resource ceiling without length limits.

10. Progressive unlocking without confounding

The useful stage order is conditional, not a fixed ten-stage ritual:

```
Exact source + data + verifier qualification
    -> short-target SFT only if source competence is insufficient
    -> one-factor curriculum / preservation comparison
    -> RL only when useful verifier-positive signal exists
    -> modular skill or bounded environment transfer
    -> frozen confirmation and scale decision
```

A stage must bind objective, eligible parent groups, trainable inventory, context, tokenizer, optimizer, budget, entry and exit tests, checkpoint policy and rollback conditions. Data access, mixture weights, maximum sequence length, LoRA rank and layer scope are independent variables. PRLC currently changes data access and proportions while keeping scope fixed. That is a defensible first study, not an omission to remove immediately. [R1]

Unlock more layers only when the current scope demonstrably underfits verified targets, the error is not a data/runtime issue, and a matched broader-scope pilot gives better held-out performance. New trainable parameters require an explicit optimizer-state policy, learning-rate warmup and frozen-reference continuity. Increasing rank also requires a specified initialization and preservation of existing factors. Do not carry an old qualification receipt across such changes.

For context expansion, register a separate length curriculum and monitor actual completion truncation. Do not truncate a verified solution and keep its correctness label. A longer source task may need a new task decomposition, not simply a larger token cap.

11. Minimum decisive training experiment

11.1 Question and design

PROPOSED experiment E1: Does a short, verified task set produce transferable acquisition in one precisely bound Rineya student, using fixed-scope LoRA and preservation replay, compared with the unchanged base?

This first run tests **whether the proposed learning setup works at all**. It does not claim that MiMo, a curriculum or RL is superior. Establishing basic acquisition avoids spending large rollout budgets on a student/runtime combination that cannot yet learn the bridge tasks reliably.

Use an original 4,096-parent acquisition bank, with 1,024 parents each for bounded integer arithmetic, symbolic variable updates, finite-set queries and instruction-constrained text transformations. Add 1,024 separate retention-training rows from training-safe preservation material; these must not be the scored preservation bank. Select unique parent families by a frozen seeded rule after decontamination. Keep all tasks text-only, with unambiguous exact answers. A teacher is optional when an independent program can produce and verify the target. These counts are a new proposal; they do not imply that the existing MaxRL bank or raw MiMo already supplies the exact requested subset.

The first target is answer-only or a tightly defined short answer envelope. Do not train unconstrained rationale outputs through the current exact-answer verifier. A later rationale-vs-answer experiment needs a new qualified envelope and independently checked explanations.

11.2 Exact proposed configuration

```
experiment_id: RINEYA-BRIDGE-SFT-E1
status: PROPOSAL_NOT_A_LAUNCH_CONFIG
base_checkpoint: null  # bind exact local Rineya ~344M export
base_weight_sha256: null
tokenizer_and_template_sha256: null
runtime_and_loader_receipt: null
native_training_authorized: false

acquisition_train_parents: 4096
retention_train_rows: 1024
selection_seed: 20260927
training_seeds: [101, 211, 307]
acquisition_check_parents: 512
retention_check_parents: 512
final_banks: separately generated; unopened during E1

objective: completion-only supervised cross-entropy
retention_mix: 1 retention sequence per 4 acquisition sequences
loss_reduction: mean per completed target, then mean across sequences
trainable_scope: fixed rank-8 LoRA on qualified q_proj/v_proj modules
lora_alpha: 16
lora_dropout: 0
base_parameters: frozen; no automatic unfreezing
precision: BF16 if native qualification passes; otherwise new plan
optimizer: AdamW on adapters only
learning_rate: 0.0001
betas: [0.9, 0.999]
epsilon: 0.00000001
weight_decay: 0.0
max_gradient_norm: 1.0
schedule: linear warmup 5% of optimizer steps, then linear decay
max_total_context: 1024
```

```
max_target_tokens: 128
truncation: reject/replace before freezing the manifest; never truncate targets
microbatch_sequences: 1
gradient_accumulation_sequences: 20  # 16 acquisition + 4 replay
training_exposure: one complete frozen sequence epoch
optimizer_steps_per_seed: 256  # 5120 / 20
checkpoint_interval_steps: 64
DEV_eval_steps: [0, 64, 128, 192, 256]
CHECK_eval: unchanged base and final step 256 only
CHECK_access: after all confirmation checkpoints are frozen
first_seed_screen: DEV only; never use CHECK to decide continuation
checkpoint_selection: final step, not best CHECK
max_nonpadding_training_tokens_per_seed: 5242880
max_native_wall_time_hours_per_seed: 24  # safety ceiling, not runtime estimate

primary_outcome: paired CHECK acquisition pass@1 change
preservation: per-family change and source-correct losses
format_and_EOS_failures: count every attempt
```

The adapter scale and optimizer numbers are conservative proposed starting values, not literature-derived optima or current repository defaults. E1 requires a standalone registered SFT route or a carefully reviewed controller extension: PRLC's existing teacher bootstrap is conditional on its own state and is not an arbitrary unrestricted SFT command. No runnable command is fabricated here. The local binding fields stay null until actual host evidence exists.

11.3 Execution gates and decision thresholds

Before E1, qualify 32 fixed non-evaluation prompts across short/near-cap lengths; compare source and zero-adapter behavior, raw sampled-token/teacher-forced logprobs, first EOS inclusion, PAD=EOS cases, one causal shift and actual trainable inventory. Freeze dtype-appropriate tolerances prospectively; the repository's numerical threshold must not be widened after seeing a failure. Measure real forward/backward memory and deterministic resume under a separately authorized one-update fixture.

Stop on unbound identities, dataset/parent collisions, unsupported task capabilities, nonfinite loss/gradients, gradient leakage outside the allowed scope, verifier inconsistency, hash drift, resource-guard breach or inconsistent resume. Keep the failure receipt and spent budget. Do not retry automatically with looser memory or statistical thresholds.

For the one-seed **operational screen on DEV only**, a proposed promising result is acquisition gain of at least 3 percentage points, no major per-family preservation regression, and no material malformed-response increase. Keep CHECK closed until the decision to run the additional seeds and all final checkpoints are frozen. This is a screening rule, not a confidence statement. A neutral result is informative: inspect training fit and error categories before adding RL.

Only then run the two other seeds under a fresh explicit budget. A stronger pilot success criterion is positive acquisition in at least two of three seeds, pooled descriptive mean gain at least 3 points, and a paired acquisition interval excluding zero under the predeclared analysis. A proposed preservation screen is no aggregate loss above 2 points and no targeted family loss above 5 points; **passing this screen does not certify noninferiority**. A later powered confirmation must establish the registered preservation margin.

If the dataset yields fewer than the required eligible parents or a different token total, stop and amend the plan before execution. Never silently fill gaps with held-out examples or shift to a different checkpoint.

11.4 What follows each outcome

Acquisition and preservation pass: E2 compares random versus easy-to-hard order using identical examples, exposures, scope and optimizer. Use fresh CHECK data because E1's results informed the decision. Only order changes; an expanding-access curriculum is a subsequent distinct comparison.

Acquisition improves but retention fails: Keep the objective and data fixed; compare replay dose and/or KL anchoring before increasing training. Identify which source-correct families fail. Do not use aggregate gains to hide those losses.

Training targets remain underfit: First inspect tokenizer, target validity, mask and LR. Only then compare q/v LoRA against a wider trainable scope at the same task/token budget. Do not infer capacity limitation from one failed run.

Training fit improves but CHECK does not: Test family/template memorization and target complexity. Prefer new parent families or better data over more epochs on the same examples.

No useful reward signal after SFT: Do not launch NGU until success or a high-G run indefinitely. Continue verified supervision or redesign tasks. Record the lack of signal as a decisive negative feasibility result.

12. NOW, NEXT, LATER, and not prioritized

NOW: Bind the exact student/runtime and a deterministic verifier; profile the selected data; complete native no-update and bounded backward/resume qualification; register E1; run only after explicit host authorization. Reuse PRLC/Q0 contracts while preserving their evidence boundaries.

NEXT: Confirm E1, isolate curriculum, then quantify preservation mechanisms. When on-policy successes and mixed groups are available, compare rejection-SFT, GRPO and MaxRL with one shared sampling/accounting contract. Separately screen per-skill LoRA against monolithic mixed LoRA before sophisticated composition.

LATER: Bounded NGU/Reinforce-Ada, score centering for measured mismatch, JumpLoRA/CLARE, Nexus, a small MiMo ReAct sandbox, and optional teacher-prompt optimization. Keep recurrent/latent architecture work alive as separately scoped science, not as a dependency for all post-training.

DO NOT PRIORITIZE: Entire-release training, full diffusion conversion, multiple simultaneous attention replacements, an unidentified GPO implementation, parameter unlocking before simple scope qualification, or any pipeline whose budget excludes failed rollouts, tool execution and reference scoring.

13. Open research questions and limits

The actual tokenizer-specific task distribution and full parent-family structure remain unknown. No native throughput or memory fit was measured. The exact local 344M checkpoint, its model path and hashes are not bound. This audit did not independently rerun repository tests or validate their reported artifacts. Most recent papers were assessed from primary abstracts and selected HTML/project material, not reproduced end to end; search-only access is explicitly recorded for CLARE and LoopSpec. No claim is made that the reviewed method set exhausts all literature through September 2026.

GPO's identity remains unresolved despite targeted searches and history retrieval; ScaleRL is a useful distinct scaling reference, not a substitute. Some named research branches were enumerated but not inspected internally. The current report is therefore an evidence-bounded research and experiment-design deliverable, not a certificate that the entire repository or dataset is qualified.

The central open question is empirical and tractable: how much useful new task competence can a fixed compact base acquire per verified training token without unacceptable source-correct losses? The staged program turns that question into a sequence of measurable decisions instead of a single speculative training run.

Appendix A. Repository evidence and audit scope

The complete observed 35-ref inventory, full SHAs and inspection-depth labels are supplied in `supporting_materials/repository_branch_inventory.csv`. The following are the major directly reviewed evidence anchors. A short SHA is only a reading aid; all references use full immutable SHAs.

Anchor	Head	Inspected material
main	`b0ca54bdc1b6`	Default head; PRLC base integration record read
progressive-rl-curriculum-v1	`5517361ec09e`	README, VALIDATION, objective source, reconnaissance read
maxrl-comp002-pretraining-qualification-v1	`21532b765269`	Directory and Q0 qualification document read
rineya-composable-subspace-stack-v1	`268eb97520f2`	README read; CPU/synthetic boundary explicit
titan-model-soup-mini-v1	`816a1e8f4d1a`	README read; native collector closed
rola-ngu-publication-reconciliation-v1	`a99f3bc41955`	Factorial README read; logical-async CPU only

The API snapshot was not a timestamp-sorted recent-activity audit. Main's integration record describes only recurrence and long-context experiment directories at the PRLC base; it does not imply the other research branches are merged. No recent-branch authorizations or resource grants were transferred. A 404 on a guessed README was treated as a missing path, not as proof of missing implementation. [R0,R4]

A.1 Implementation-specific integration hazards

PRLC's source is not a reproduction of MaxRL, NGU, DAPO or MiMo's full recipe. It uses a CPU reference model, fixed rank-8 LoRA, no top-k/top-p sampling truncation, explicit EOS masks and an intentionally narrow single-turn task contract. Its schema disallows sealed/test labels and its native collector does not advertise MiMo domain capabilities. Those are safeguards to preserve in any extension. [R1–R3]

Q0 records source/data integrity and separately completed test suites, not a single new full native/data resolve. It binds candidate R/S banks but leaves checkpoint, exact native scope, margins and compute budgets open. Its exported package hashes must not be invalidated by adding utilities inside a recursively hashed directory. [R7]

CSS is a one-explicit-scope prototype with protected, shared/private and compression factors. Its example scope is one q-projection matrix at layer 16, rank 16, with Titan-Sparse/Opallo weights 0.5/0.5. That example cannot be applied unchanged to a student with fewer than 17 layers, different parameter names, or a different architecture. Native ancestry, architecture and exact scope are hard prerequisites, not implementation details to infer by name. [R5]

Appendix B. Assessments of all 32 brainstormed methods

Resource profiles are planning estimates and qualitative constraints; no per-method native GPU-hour/VRAM benchmark was run. Published evidence describes the stated paper setting. Benefits, experimental comparisons and falsifiers below are proposed Rineya research judgments. The CSV includes one record per method with independent fields for filtering and editing.

B01. WSD

Track: Optimization schedule. **Priority:** NEXT. **Sources:** [P01].

Mechanism and evidence. Warm up learning rate, maintain a stable plateau, then decay for consolidation. It changes step size, not supervision or architecture. A published small-model training strategy; no measured WSD advantage on this MiMo/Rineya setup.

Assumptions and expected benefit. A meaningful stable phase fits the budget; compare identical tokens and optimizer scope. A very short pilot may barely distinguish schedules. Proposed upside: Simpler continuation and planned stopping points; potentially better consolidation.

Failure and preservation risks. Too much plateau learning can erase old behavior; apparent gains may come only from different final learning rates. Forgetting: Medium; controlled by data, scope and step size, not by the scheduler name. Instability: Medium during the stable phase or resumed decay.

Implementation and data. Complexity: Low. Needs: Verified demonstrations or RL tasks, depending on the objective. Repository boundary: Historical project context; not audited as a standalone current branch.

Resources (profile A). Compute: One SFT-like training pass per token; LoRA still backpropagates through the frozen network. VRAM: 344M BF16 base is about 0.64 GiB; activations/logits/runtime dominate the remaining envelope. 8 GiB fit is plausible, unmeasured. Host RAM: Stream small batches; preserve existing admission guards. Avoid full dataset materialization. Storage: Base weights plus small adapters, optimizer/RNG checkpoints and bounded token caches; actual shapes required.

Test and interaction. Match learning-rate peak, token budget, endpoint evaluations and checkpoint-selection rules. Compatible with LoRA and replay; test separately from curriculum and unlocking.

Falsifier / stop signal. Fails priority if equal-budget cosine or warmup-decay matches or beats it on capability/retention.

B02. LoRA

Track: Parameter-efficient training. **Priority:** NOW. **Sources:** [P02,R1].

Mechanism and evidence. Freeze base W and learn a scaled low-rank update ΔW=(alpha/r)BA. Parameter count is r(d_in+d_out) per adapted matrix. Foundational method; PRLC implements rank-8 q/v targets but has only CPU qualification.

Assumptions and expected benefit. Chosen modules carry the required adaptation; low rank may restrict representational change. Exact GQA shapes and tied parameters must be inventoried. Proposed upside: Lower optimizer-state cost and clean adapter checkpoints; strong simple baseline.

Failure and preservation risks. Rank/module bottleneck; low training loss without novel problem solving; adapter composition can interfere. Forgetting: Medium; freezing the base does not freeze the model function. Instability: Low to medium; watch scale, LR and initially inactive factors.

Implementation and data. Complexity: Low to medium. Needs: Short, verified response examples; disjoint general replay. Repository boundary: IMPLEMENTED + SYNTHETIC-ONLY in PRLC; native fit unqualified.

Resources (profile A). Compute: One SFT-like training pass per token; LoRA still backpropagates through the frozen network. VRAM: 344M BF16 base is about 0.64 GiB; activations/logits/runtime dominate the remaining envelope. 8 GiB fit is plausible, unmeasured. Host RAM: Stream small batches; preserve existing admission guards. Avoid full dataset materialization. Storage: Base weights plus small adapters, optimizer/RNG checkpoints and bounded token caches; actual shapes required.

Test and interaction. Native zero-adapter parity, trainable inventory, finite gradients, held-out capability and source-correct losses. Natural with SFT/GRPO; per-skill adapters require dose and routing controls.

Falsifier / stop signal. Increase scope only if verified targets remain underfit and full/alternate-scope controls improve held-out transfer.

B03. MaxRL

Track: RL objective. **Priority:** NEXT. **Sources:** [P03,R7].

Mechanism and evidence. Targets log probability of correctness across prompts rather than only mean success probability; finite-compute estimators approximate a maximum-likelihood RL objective. Published objective and experiments; local Q0 source/data binding exists, but it provides no native runner or training authorization.

Assumptions and expected benefit. Correctness must be observable and some successes discoverable. Estimator normalization, sampling and reward-group conventions must match the selected version. Proposed upside: Rebalance learning away from already-easy problems, potentially improve difficult-task success.

Failure and preservation risks. All-failure groups remain uninformative; biased success-only curation can misrepresent the estimator. Forgetting: Medium to high without independent preservation loss. Instability: Medium to high for rare positives, weighting extremes and stale policies.

Implementation and data. Complexity: Medium. Needs: Deterministically verifiable prompts and raw independent rollout records. Repository boundary: SOURCE/DATA PASS, NATIVE BLOCKED; 462 original and 46 new tests separately reported.

Resources (profile B). Compute: Generation + policy scoring/backward + reference scoring + verification + evaluation; often far more than SFT. VRAM: Model, activations, logits and rollout KV; serialize trajectories on 8 GiB. CPU reference exchanges VRAM for RAM/time. Host RAM: Frozen CPU reference and trajectory buffers can be binding; bounded queues required. Storage: Store raw token IDs, behavior logprobs, rewards, policy versions, checkpoints and spending ledgers.

Test and interaction. Equal total generation budgets; pass@1, mixed-group rate, per-difficulty gain, all discarded attempts counted. LoRA and replay compatible; do not silently stack NGU or Reinforce-Ada allocation changes.

Falsifier / stop signal. No advantage over matched GRPO/RSFT after counting all sampling and retention cost.

B04. RoLA

Track: Attention architecture. **Priority:** LATER. **Sources:** [P04,R8].

Mechanism and evidence. Apply rotary positions after a nonlinear low-rank feature map to support global linear-attention summaries; original work targets video diffusion transformers. Cross-domain paper. Rineya branch implements an inspired causal residual beside dense attention, not a paper-exact sparse replacement.

Assumptions and expected benefit. Causal language use, native rotary pairs and activation scale need new validation. Zero residual gate can delay learning of inner projections. Proposed upside: Potential added recurrent summary capacity; not automatic speedup while dense attention remains.

Failure and preservation risks. Distribution shift, faulty causality/cache, inactive gate or quality loss. Forgetting: High for replacement; residual adaptation may reduce but not remove risk. Instability: High: new recurrent states, normalization and gating.

Implementation and data. Complexity: High. Needs: Language adaptation targets plus long-context and causal perturbation tests. Repository boundary: IMPLEMENTED + SYNTHETIC-ONLY causal adaptation; no pretrained launch.

Resources (profile C). Compute: Additional model visits, objective changes or conversion training; no defensible fixed multiplier without prototype profiling. VRAM: Recurrent activations/new states can dominate. Checkpointing changes time/memory; 24–48 GiB may be operationally easier, not guaranteed necessary. Host RAM: Model/teacher state and preprocessing can exceed constrained host RAM. Storage: Separate architecture checkpoints and exact runtime/config artifacts; do not overwrite the base.

Test and interaction. Zero/active-path parity, future-token perturbations, cache agreement, matched added-MLP control. Factorial with NGU exists, but keep pruning separate; not equivalent to LoRA.

Falsifier / stop signal. No useful gain over equal-parameter dense residual or unacceptable added latency.

B05. Delta weights / task vectors

Track: Composition and retention. **Priority:** NEXT. **Sources:** [R5,R6].

Mechanism and evidence. Express a child as W0+ΔWi and compose common-base updates with frozen coefficients; sparsification or projection changes the update, not necessarily the execution graph. Source-backed mechanics and inherited trade-off observations; low cosine is not behavioral independence.

Assumptions and expected benefit. Exact common base, tokenizer, tensor alignment and complete ancestry; BF16-rounded deltas cannot be reconstructed by casting. Proposed upside: Cheap post-hoc controls, attribution and potentially useful skill composition.

Failure and preservation risks. Double-count inherited updates, compare unmatched norms, average raw LoRA factors or combine incompatible models. Forgetting: High unless functionally measured; small norm is not safe by itself. Instability: No optimizer in merge-only arm; numerical/casting hazards remain.

Implementation and data. Complexity: Medium. Needs: Fresh donor-skill evaluation plus supplied ancestor training ledgers. Repository boundary: SOUP/CSS builder mechanics only; separate 596,049,920-parameter family scope.

Resources (profile D). Compute: Merge-only: tensor reads/writes and possibly SVD; Fisher/tangent variants additionally need backward/JVP. VRAM: CPU streaming can avoid GPU allocation for simple merges; calibration backward is a separate fit test. Host RAM: One artifact/tensor slice at a time; SVD workspace and page cache must be measured. Storage: 596M FP32 output is about 2.22 GiB tensor data; build selected candidates serially, not all arms.

Test and interaction. Base, endpoints, dose-matched single deltas, sum/average and norm controls on frozen CAL/CHECK. Complementary to per-skill LoRA; projection, pruning and dose are distinct factors.

Falsifier / stop signal. Composition does not outperform the best component at matched dose/compute or loses protected source-correct items.

B06. DPO

Track: Preference optimization. **Priority:** NEXT. **Sources:** [P05].

Mechanism and evidence. Optimize a logistic margin between chosen and rejected response log-ratios relative to a reference policy. It requires preference comparisons, not just tasks. Established pairwise objective; no suitable pairs established in the inspected MiMo task schema.

Assumptions and expected benefit. Pairs share a prompt; verifier/judge can defend the ordering; length and formatting cannot stand in for correctness. Proposed upside: Correct common mistakes without an online policy-gradient loop.

Failure and preservation risks. Chosen/rejected length confounding, weak negatives, incorrect judges, preference overfitting. Forgetting: Medium; a reference term is not a complete retention constraint. Instability: Medium with large beta or poorly separated response probabilities.

Implementation and data. Complexity: Medium. Needs: Audited same-prompt positive/negative trajectories, generated separately if absent. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile P). Compute: Train on both chosen and rejected response tokens, plus frozen-reference scoring. Cache reference logprobs only with immutable model/tokenizer/template identities; count pair generation separately. VRAM: Policy activations and logits remain; a co-resident reference adds memory. Precomputed reference scores avoid simultaneous loading. Exact 8 GiB fit is unmeasured. Host RAM: Stream pairs and bounded reference-score caches; CPU reference scoring exchanges VRAM for RAM and time. Storage: Base/adapters, optimizer/RNG state, pair provenance and reference-logprob cache.

Test and interaction. Compare SFT on positives against DPO with equal source prompts and total target tokens; report pair construction cost. Alternative to the first RL stage, not a mandatory pre-RL step.

Falsifier / stop signal. No incremental held-out correctness over positive-only SFT; gains vanish on length-matched pairs.

B07. Recurrence

Track: Adaptive computation architecture. **Priority:** LATER. **Sources:** [P14,R0].

Mechanism and evidence. Reuse physical blocks or latent update functions for multiple visits, expanding computation without equally expanding stored weights. Published looped models; current repository includes recurrence-related branches, not proof every Rineya checkpoint benefits.

Assumptions and expected benefit. Reused blocks can process revisited states; gradients, positions, caches and residual scaling remain stable. Proposed upside: More depth per parameter, possibly adaptive reasoning effort.

Failure and preservation risks. Repeated errors, fixed-point collapse, almost-zero learned gates, high serial latency. Forgetting: High for retrofits; baseline behavior needs explicit protection. Instability: High; tied gradients and visit alignment change optimization.

Implementation and data. Complexity: High. Needs: Step-sensitive tasks with transfer tests and matched nonrecurrent adaptation. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile C). Compute: Additional model visits, objective changes or conversion training; no defensible fixed multiplier without prototype profiling. VRAM: Recurrent activations/new states can dominate. Checkpointing changes time/memory; 24–48 GiB may be operationally easier, not guaranteed necessary. Host RAM: Model/teacher state and preprocessing can exceed constrained host RAM. Storage: Separate architecture checkpoints and exact runtime/config artifacts; do not overwrite the base.

Test and interaction. Matched training/inference compute, zero-loop and one-loop baselines, gate/gradient diagnostics. Dependency for DeepLoop/LoopSpec; do not combine three recurrence variants before a base effect exists.

Falsifier / stop signal. No capability gain beyond an equally expensive nonrecurrent control or only longer answers.

B08. CLARE

Track: Sparse continual learning. **Priority:** NEXT / bounded study. **Sources:** [P06,P35].

Mechanism and evidence. The intended class-incremental method learns sparsity-based task masks, then constrains updates in a shared adapter space. A VLA routing paper uses the same name. Primary abstract supports a class-incremental framework, not a direct guarantee about autoregressive LLM reasoning.

Assumptions and expected benefit. Learned masks isolate meaningful skills rather than merely reducing update size; text-domain adaptation differs from the original setting. Proposed upside: Potential selective skill acquisition with less interference.

Failure and preservation risks. Mask saturation, cross-skill conflicts and misleading gains from attenuation alone. Forgetting: Medium to high; test per-skill acquisition and retention jointly. Instability: Medium for learned hard gates and sparsity penalties.

Implementation and data. Complexity: Medium to high. Needs: Two demonstrably distinct, independently verifiable skill sets and retention anchors. Repository boundary: Two titan-clare branch heads enumerated; contents not audited here.

Resources (profile A). Compute: One SFT-like training pass per token; LoRA still backpropagates through the frozen network. VRAM: 344M BF16 base is about 0.64 GiB; activations/logits/runtime dominate the remaining envelope. 8 GiB fit is plausible, unmeasured. Host RAM: Stream small batches; preserve existing admission guards. Avoid full dataset materialization. Storage: Base weights plus small adapters, optimizer/RNG checkpoints and bounded token caches; actual shapes required.

Test and interaction. Dense LoRA versus sparse/masked LoRA, same capacity/dose controls; sequential and simultaneous acquisition distinguished. Close to JumpLoRA and isolation; run competing mechanisms before stacking them.

Falsifier / stop signal. No retention/acquisition Pareto improvement over sparse magnitude/dose-matched controls.

B09. Dream-RSI

Track: Agent exploration orchestration. **Priority:** LATER / teacher tooling. **Sources:** [P07].

Mechanism and evidence. Optimize how an agent branches, allocates parallel exploration and stops, using evolving discovery worlds and replay of recorded search trees. Published framework changes exploration control; it is not evidence of autonomous weight improvement in the underlying solver.

Assumptions and expected benefit. Recorded alternatives represent deployment conditions; unseen counterfactual branches cannot be scored for free. Proposed upside: Better verified teacher-trajectory discovery per environment call.

Failure and preservation risks. Simulator overfit, evaluator leakage and large collection costs hidden behind cheap replay. Forgetting: Not applicable to frozen solver weights; downstream distilled data can still cause forgetting. Instability: Controller instability and nonstationary rewards rather than ordinary model-gradient instability.

Implementation and data. Complexity: High. Needs: Audited search trees with all actions, costs and outcomes. Repository boundary: Dream-RSI branch enumerated only.

Resources (profile E). Compute: Teacher/agent calls, tool execution, judging and search dominate; frozen policy alone does not make the workflow cheap. VRAM: Local solver fit is separate; external solver memory is not part of the 8 GiB host but its monetary cost must be counted. Host RAM: Sandbox/container processes and artifacts may dominate. Storage: Environment images and outputs are additional to dataset parquet; source descriptors do not bound total storage.

Test and interaction. Fresh online worlds, identical solver, fixed environment/teacher budget, no replayed-world final test. Potential upstream of SFT; overlaps Combee/ReAct at harness level rather than low-rank training.

Falsifier / stop signal. Replay gains fail to transfer to new online tasks or cost more per verified solution.

B10. JumpLoRA

Track: Sparse adapters. **Priority:** NEXT. **Sources:** [P08].

Mechanism and evidence. Use learned JumpReLU-style sparse adapter activation/update structure for continual parameter isolation rather than plain dense low-rank updates. LLM continual-learning evidence is more directly relevant than vision-only isolation, but no native Rineya result was verified.

Assumptions and expected benefit. Learned support remains useful across tasks and does not saturate; implementation must reproduce the actual gating/regularization convention. Proposed upside: Potentially stronger modularity and preservation than plain LoRA.

Failure and preservation risks. Dead gates, task fragmentation, redundant sparse modules and dense-kernel overhead. Forgetting: Medium; isolation limits some interference, not all shared-state effects. Instability: Medium to high near gating thresholds.

Implementation and data. Complexity: Medium to high. Needs: Two skill streams, separate replay/evaluation, fixed training-order policy. Repository boundary: JumpLoRA branch enumerated only.

Resources (profile A). Compute: One SFT-like training pass per token; LoRA still backpropagates through the frozen network. VRAM: 344M BF16 base is about 0.64 GiB; activations/logits/runtime dominate the remaining envelope. 8 GiB fit is plausible, unmeasured. Host RAM: Stream small batches; preserve existing admission guards. Avoid full dataset materialization. Storage: Base weights plus small adapters, optimizer/RNG checkpoints and bounded token caches; actual shapes required.

Test and interaction. Dense LoRA, magnitude-sparse and learned-gate controls at equal budgets; support overlap plus functional losses. Alternative to CLARE; compose only after each isolated adapter actually acquires its skill.

Falsifier / stop signal. No advantage over dense or simpler sparse adaptation after accounting for trainable count and norm.

B11. GFlowRL

Track: Distribution-matching RL. **Priority:** LATER. **Sources:** [P09].

Mechanism and evidence. Train toward a reward-tilted, reference-anchored trajectory distribution, using group-based normalization and off-policy corrections rather than only optimizing expected reward. Published large-model reasoning experiments; no demonstrated transfer to a 344M student in this audit.

Assumptions and expected benefit. Reference support includes useful solutions and reward scaling is meaningful. A changed sampler must retain the correct behavior probabilities. Proposed upside: Preserve useful solution diversity while increasing verified quality.

Failure and preservation risks. Distribution collapse, unreliable normalization or expensive sampling for small-student rare successes. Forgetting: Medium to high; reference anchoring is not a task-bank retention guarantee. Instability: High relative to baseline GRPO; flow gaps and off-policy clipping require qualification.

Implementation and data. Complexity: High. Needs: Verifiable trajectories, reference likelihoods and full cost records. Repository boundary: GFlowRL branch enumerated only.

Resources (profile B). Compute: Generation + policy scoring/backward + reference scoring + verification + evaluation; often far more than SFT. VRAM: Model, activations, logits and rollout KV; serialize trajectories on 8 GiB. CPU reference exchanges VRAM for RAM/time. Host RAM: Frozen CPU reference and trajectory buffers can be binding; bounded queues required. Storage: Store raw token IDs, behavior logprobs, rewards, policy versions, checkpoints and spending ledgers.

Test and interaction. Compare quality/diversity at matched generation budget, not only pass@many. Alternative objective to MaxRL/GRPO; avoid combining losses without a derivation.

Falsifier / stop signal. No pass@1 or diversity-efficiency improvement beyond matched baselines; benefits require unaffordable rollouts.

B12. Deja Vu

Track: Inference contextual sparsity. **Priority:** LATER / systems. **Sources:** [P10].

Mechanism and evidence. Predict input-dependent active attention heads and MLP units, then execute sparse work with hardware-aware scheduling. Original large-model inference evidence; this is not recurrence or a training curriculum.

Assumptions and expected benefit. Prediction overhead is smaller than avoided compute on the actual compact model and consumer GPU. Proposed upside: Lower inference cost may fund more evaluation or rollout samples.

Failure and preservation risks. Small matrices underutilize hardware; prediction mistakes degrade outputs; sparse zeros alone yield no speed. Forgetting: No parameter learning required; approximate inference can nevertheless lose capabilities. Instability: Not an optimizer intervention.

Implementation and data. Complexity: High. Needs: Representative prompts for predictor/calibration, kept separate from final test. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile F). Compute: Inference engineering: measure all predictor/draft/replay work and concurrency. VRAM: Intended cache/work savings are host- and workload-dependent; cannot infer them from paper percentages. Host RAM: Runtime buffers, retained caches and host offload need explicit ceilings. Storage: Profiling traces and separate runtime versions; no new full checkpoint inherently required.

Test and interaction. Actual wall-time, memory and output-quality comparison at batch one and rollout batch sizes. Could assist rollout generation only after policy/logprob parity is qualified.

Falsifier / stop signal. No end-to-end speedup on the exact host or unacceptable deviation from dense behavior.

B13. F-WANDA

Track: Post-training pruning. **Priority:** LATER / bounded delta screen. **Sources:** [P11].

Mechanism and evidence. Allocate per-row keep budgets using empirical Fisher information, supplementing WANDA calibration. The paper prunes model weights; pruning deltas is a distinct adaptation. Published one-shot pruning evidence on larger models; the Rineya delta-retention hypothesis is extrapolation.

Assumptions and expected benefit. Fisher estimates generalize beyond calibration; preserving sensitive rows helps the specific functional objectives. Proposed upside: Potentially better budget allocation than uniform or magnitude-only pruning.

Failure and preservation risks. Noisy Fisher, calibration overfit, deleting skill-bearing changes, confusing full weights with deltas. Forgetting: High unless independently measured. Instability: No optimizer update needed for pruning, but backward Fisher accumulation must be finite and memory-safe.

Implementation and data. Complexity: Medium. Needs: Calibration inputs and losses plus untouched acquisition/retention checks. Repository boundary: F-WANDA branch exists; guessed README was unavailable, so contents remain unaudited.

Resources (profile D). Compute: Merge-only: tensor reads/writes and possibly SVD; Fisher/tangent variants additionally need backward/JVP. VRAM: CPU streaming can avoid GPU allocation for simple merges; calibration backward is a separate fit test. Host RAM: One artifact/tensor slice at a time; SVD workspace and page cache must be measured. Storage: 596M FP32 output is about 2.22 GiB tensor data; build selected candidates serially, not all arms.

Test and interaction. Same retained count per model, magnitude/WANDA/Fisher row allocation, norm-matched controls. Post-hoc factor after training; not a simultaneous change in the first curriculum trial.

Falsifier / stop signal. Fails to improve retention/acquisition trade-off beyond simpler matched sparsification.

B14. SoftToken

Track: Continuous reasoning training. **Priority:** LATER. **Sources:** [P12].

Mechanism and evidence. Learn continuous CoT using token-mixture embeddings with injected noise for RL exploration; the paper also studies discrete-token deployment after soft training. Math experiments with Llama/Qwen models up to 8B; pass@many and retention observations do not establish 344M success.

Assumptions and expected benefit. Soft-state training and discrete deployment are mutually compatible; noise/logprob semantics are correct. Proposed upside: Potential diversity and gentler behavioral drift.

Failure and preservation risks. Embedding mixtures are out of distribution; continuous states are not automatically parallel logical branches. Forgetting: Medium, empirical rather than guaranteed preservation. Instability: High compared with standard discrete rollout training.

Implementation and data. Complexity: High. Needs: Verifiable tasks and a defined continuous sampling/reward path. Repository boundary: Soft-token branch enumerated only.

Resources (profile C). Compute: Additional model visits, objective changes or conversion training; no defensible fixed multiplier without prototype profiling. VRAM: Recurrent activations/new states can dominate. Checkpointing changes time/memory; 24–48 GiB may be operationally easier, not guaranteed necessary. Host RAM: Model/teacher state and preprocessing can exceed constrained host RAM. Storage: Separate architecture checkpoints and exact runtime/config artifacts; do not overwrite the base.

Test and interaction. Discrete versus soft training with equal effective compute; test discrete and soft deployment separately. Not a drop-in MiMo text field; separate from Looped Flow and diffusion.

Falsifier / stop signal. No improvement when evaluated under the intended deployable discrete decoder and matched compute.

B15. Looped Flow

Track: Latent flow reasoning. **Priority:** LATER. **Sources:** [P13].

Mechanism and evidence. Train recurrent latent refinement with local denoising and a changing noise schedule; inference integrates learned updates over a chosen trajectory. Structured reasoning evidence; applying it to frozen CED/Rineya features is a new hypothesis, not the full published model.

Assumptions and expected benefit. Latent states encode useful solutions and local objectives generalize to language outputs. Proposed upside: Iterative correction without long explicit verbal traces.

Failure and preservation risks. Noise schedule mismatch, converging to wrong states, latent improvements that do not decode to correct answers. Forgetting: High for architectural modification; lower scope does not prove protection. Instability: High: training/inference dynamical mismatch.

Implementation and data. Complexity: High. Needs: Supervised/verifiable structured tasks, target-latent and decoder contracts. Repository boundary: CED paired-host branch enumerated; no new native evidence established.

Resources (profile C). Compute: Additional model visits, objective changes or conversion training; no defensible fixed multiplier without prototype profiling. VRAM: Recurrent activations/new states can dominate. Checkpointing changes time/memory; 24–48 GiB may be operationally easier, not guaranteed necessary. Host RAM: Model/teacher state and preprocessing can exceed constrained host RAM. Storage: Separate architecture checkpoints and exact runtime/config artifacts; do not overwrite the base.

Test and interaction. Ordinary residual/MLP versus flow refinement at matched iterations, tokens and trainable count. Alternative latent-compute route to EqR/recurrence, not an automatic compatible stack.

Falsifier / stop signal. No gain beyond a deterministic recurrent residual control or unstable decode under longer integration.

B16. DeepLoop

Track: Looped-model stabilization. **Priority:** LATER / recurrence dependency. **Sources:** [P14].

Mechanism and evidence. Account for repeated visits to tied parameters in Post-LN residual scaling; the conservative aligned regime changes the depth exponent relative to ordinary DeepNorm. Published GPT-scale looped-language results and first-order analysis. It is not a general instruction to rescale every pretrained pre-LN model.

Assumptions and expected benefit. Target architecture matches the normalization and initialization assumptions; effective depth and visit alignment are tracked. Proposed upside: Better trainability when genuine recurrence is activated.

Failure and preservation risks. Importing initialization rules into an already-trained network changes its function; hidden norm drift. Forgetting: High for naive retrofit. Instability: Designed to address instability, but outside-assumption use may create it.

Implementation and data. Complexity: High. Needs: Language or structured-task looped-model training and a native recurrent baseline. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile C). Compute: Additional model visits, objective changes or conversion training; no defensible fixed multiplier without prototype profiling. VRAM: Recurrent activations/new states can dominate. Checkpointing changes time/memory; 24–48 GiB may be operationally easier, not guaranteed necessary. Host RAM: Model/teacher state and preprocessing can exceed constrained host RAM. Storage: Separate architecture checkpoints and exact runtime/config artifacts; do not overwrite the base.

Test and interaction. No-loop parity, loop-count sweeps, gradient norms and equal-training-compute accuracy. Supports recurrence; not independent evidence that more looping helps a given student.

Falsifier / stop signal. No stability or held-out benefit on a matched looped model, or required retraining exceeds project value.

B17. Discrete Diffusion

Track: Different generative objective. **Priority:** DO NOT PRIORITIZE. **Sources:** [P15].

Mechanism and evidence. Predict masked/corrupted discrete tokens through a denoising process rather than ordinary left-to-right next-token generation. Published language diffusion models support the paradigm; not cheap proof that an existing autoregressive Rineya can be switched by a loss flag.

Assumptions and expected benefit. Architecture, attention masking, corruption schedule and deployment decoder support the new objective. Proposed upside: Parallel/refinement decoding and alternative learning signals.

Failure and preservation risks. Pretrained distribution mismatch, expensive conversion, many denoising steps and verifier ambiguity. Forgetting: High during conversion. Instability: High without a well-qualified objective and schedule.

Implementation and data. Complexity: Very high. Needs: Large appropriate denoising-training corpus, not environment descriptors alone. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile C). Compute: Additional model visits, objective changes or conversion training; no defensible fixed multiplier without prototype profiling. VRAM: Recurrent activations/new states can dominate. Checkpointing changes time/memory; 24–48 GiB may be operationally easier, not guaranteed necessary. Host RAM: Model/teacher state and preprocessing can exceed constrained host RAM. Storage: Separate architecture checkpoints and exact runtime/config artifacts; do not overwrite the base.

Test and interaction. AR baseline with matched pretraining/post-training FLOPs and actual serving cost. Conflicts with treating the current AR runtime and rollout likelihoods as unchanged.

Falsifier / stop signal. No better quality/latency frontier after counting conversion and sampling cost.

B18. Full-Bandwidth Transformer

Track: Cross-token hidden feedback. **Priority:** LATER. **Sources:** [P16].

Mechanism and evidence. Feed previous top-layer hidden information into subsequent token processing through a learned feedback pathway; training must expose the model to that pathway. Published language-model architecture, not a parameter-free inference optimization.

Assumptions and expected benefit. Feedback carries useful state and scheduled training resolves teacher-forcing/inference mismatch. Proposed upside: Transfer hidden reasoning information beyond the emitted discrete token.

Failure and preservation risks. Feedback error accumulation, unstable cache state, expensive training integration. Forgetting: High for pretrained retrofit. Instability: High; recurrent feedback scale and train/test mismatch.

Implementation and data. Complexity: High. Needs: Language examples with a tested feedback-training contract. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile C). Compute: Additional model visits, objective changes or conversion training; no defensible fixed multiplier without prototype profiling. VRAM: Recurrent activations/new states can dominate. Checkpointing changes time/memory; 24–48 GiB may be operationally easier, not guaranteed necessary. Host RAM: Model/teacher state and preprocessing can exceed constrained host RAM. Storage: Separate architecture checkpoints and exact runtime/config artifacts; do not overwrite the base.

Test and interaction. Ordinary decoder versus feedback and equal-parameter control; greedy/teacher-forced parity checks. Different from looping a block; do not merge its gains with SoftToken results.

Falsifier / stop signal. No independent quality gain after controlling token length, parameters and compute.

B19. Explorative Modeling

Track: Training-time generative exploration. **Priority:** LATER. **Sources:** [P17].

Mechanism and evidence. Use multiple training-time candidates and favor better matches, adding an exploration axis to generative learning. Published generative framework; a reasoning-LLM post-training recipe requires additional derivation.

Assumptions and expected benefit. Candidate-selection metric tracks useful generalization; K-way exploration provides information rather than easy-sample bias. Proposed upside: Potentially richer candidate exploration and supervision.

Failure and preservation risks. Winner selection overfits the scorer; discarded candidates hide compute; mode coverage may fall. Forgetting: Medium to high during adapted training. Instability: Medium to high depending on hard selection and estimator.

Implementation and data. Complexity: High. Needs: Targets or trustworthy evaluators for candidates; complete candidate logs. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile C). Compute: Additional model visits, objective changes or conversion training; no defensible fixed multiplier without prototype profiling. VRAM: Recurrent activations/new states can dominate. Checkpointing changes time/memory; 24–48 GiB may be operationally easier, not guaranteed necessary. Host RAM: Model/teacher state and preprocessing can exceed constrained host RAM. Storage: Separate architecture checkpoints and exact runtime/config artifacts; do not overwrite the base.

Test and interaction. Match total candidate compute; compare ordinary SFT/rejection and held-out diversity. Related pedagogically to rejection, not mathematically interchangeable with GRPO or MaxRL.

Falsifier / stop signal. No advantage beyond a simpler verifier-filtered baseline at identical selection cost.

B20. Equilibrium Reasoners

Track: Attractor-based latent computation. **Priority:** LATER. **Sources:** [P18].

Mechanism and evidence. Learn task-conditioned solution attractors and scale reasoning with iterative convergence or multiple initializations. Strong structured-task evidence is not general proof of natural-language reasoning or correctness certification.

Assumptions and expected benefit. Useful solutions form learnable attractors; residual convergence corresponds to a valid solution often enough to be useful. Proposed upside: Adaptive depth and potentially low-state iterative solving.

Failure and preservation risks. Wrong fixed points, convergence without correctness, large inference iteration counts. Forgetting: High for a direct language-model retrofit. Instability: High for unstable dynamics or poorly controlled implicit/truncated gradients.

Implementation and data. Complexity: Very high. Needs: Structured tasks with solution validators and explicit latent/decoder design. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile C). Compute: Additional model visits, objective changes or conversion training; no defensible fixed multiplier without prototype profiling. VRAM: Recurrent activations/new states can dominate. Checkpointing changes time/memory; 24–48 GiB may be operationally easier, not guaranteed necessary. Host RAM: Model/teacher state and preprocessing can exceed constrained host RAM. Storage: Separate architecture checkpoints and exact runtime/config artifacts; do not overwrite the base.

Test and interaction. Correctness versus residual norm, fixed iteration and adaptive stopping, independent task families. Alternative to Looped Flow; combine only after identifying distinct benefits.

Falsifier / stop signal. Converged states fail correctness tests or equal-compute ordinary recurrence performs as well.

B21. Nexus

Track: Multi-domain optimizer. **Priority:** LATER / focused comparison. **Sources:** [P19].

Mechanism and evidence. Encourage different data-source gradients to approach a common minimum, rather than only minimizing their summed loss. Published pretraining experiments span 130M–3B; transfer to narrow post-training remains unmeasured.

Assumptions and expected benefit. Domain partition is meaningful and gradient agreement aligns with both acquisition and retention. Proposed upside: More cross-domain generalization despite similar average loss.

Failure and preservation risks. Alignment may suppress useful conflicting skills; inner computations consume budget and memory. Forgetting: Medium; agreement is not a complete forgetting guarantee. Instability: Medium to high for nested updates or normalization.

Implementation and data. Complexity: High. Needs: Separately labeled domains with balanced, comparable gradient estimates. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile A+). Compute: SFT baseline plus domain/inner-gradient work; measure overhead rather than equating steps. VRAM: Potential extra gradient/model buffers; may exceed 8 GiB even when LoRA fits. Host RAM: Extra domain statistics; avoid simultaneous full state copies. Storage: Baseline state plus domain/optimizer records.

Test and interaction. AdamW versus Nexus-inspired/paper-faithful optimizer at matched FLOPs, not steps alone. Potential tension with forcing skill deltas orthogonal; gradient alignment and delta cosine are different quantities.

Falsifier / stop signal. No downstream gain at matched compute, or alignment sacrifices genuinely distinct skill performance.

B22. ReAct

Track: Agent interaction format. **Priority:** NEXT / environment pilot. **Sources:** [P20].

Mechanism and evidence. Interleave model reasoning/actions with environment observations. A harness can implement this without changing weights; training on trajectories is a separate step. Foundational agent framework; not a guarantee an undersized policy can solve full repository tasks.

Assumptions and expected benefit. Tools, observations, action schema and termination are reliable and bounded. Proposed upside: Create grounded execution evidence and verifiable trajectories.

Failure and preservation risks. Tool-output leakage, invalid actions, excessively long histories and environment-specific shortcuts. Forgetting: No weight change in prompting-only mode; distilled use has ordinary forgetting risk. Instability: Harness/credit-assignment failures rather than an inherent optimizer issue.

Implementation and data. Complexity: Medium to high. Needs: Tasks, sandbox/tool contracts and raw action-observation traces. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile E). Compute: Teacher/agent calls, tool execution, judging and search dominate; frozen policy alone does not make the workflow cheap. VRAM: Local solver fit is separate; external solver memory is not part of the 8 GiB host but its monetary cost must be counted. Host RAM: Sandbox/container processes and artifacts may dominate. Storage: Environment images and outputs are additional to dataset parquet; source descriptors do not bound total storage.

Test and interaction. Tool success, invalid-action rate, full task reward, cost and robust replay. Essential bridge for agentic MiMo; complementary to Combee/Dream as upstream controls.

Falsifier / stop signal. No reliable completion beyond simpler direct-tool baselines or trajectory cost exceeds a short-task bridge.

B23. MoBA

Track: Long-context sparse attention. **Priority:** LATER / systems-architecture. **Sources:** [P21].

Mechanism and evidence. Route queries to selected key/value blocks while retaining causal constraints, reducing attended context work. Published long-context attention method; benefits depend on sequence length and kernel implementation.

Assumptions and expected benefit. Retrieval-critical blocks are selected; compact short-context workloads amortize routing overhead. Proposed upside: Potential longer useful context under a fixed budget.

Failure and preservation risks. Missed evidence, routing discontinuities and poor short-sequence speed. Forgetting: High for unqualified replacement; no guarantee existing attention behavior survives. Instability: Medium to high if trained routing changes sharply.

Implementation and data. Complexity: High. Needs: Long-context adaptation and location-controlled retrieval evaluations. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile C). Compute: Additional model visits, objective changes or conversion training; no defensible fixed multiplier without prototype profiling. VRAM: Recurrent activations/new states can dominate. Checkpointing changes time/memory; 24–48 GiB may be operationally easier, not guaranteed necessary. Host RAM: Model/teacher state and preprocessing can exceed constrained host RAM. Storage: Separate architecture checkpoints and exact runtime/config artifacts; do not overwrite the base.

Test and interaction. Dense baseline, same visible context, retrieval failures by evidence location, native speed. An alternative attention design to NSA/RoLA, not three add-ons to enable simultaneously.

Falsifier / stop signal. No long-context quality/compute improvement or critical retrieval degradation.

B24. Combee

Track: Prompt optimization. **Priority:** NEXT / teacher tooling. **Sources:** [P22].

Mechanism and evidence. Learn/improve agent prompts by scanning experience with controlled parallelism and batching; the agent weights need not change. Published prompt-learning framework; benefits are harness-specific and do not demonstrate weight distillation.

Assumptions and expected benefit. Trace feedback is trustworthy and training tasks are distinct from prompt-evaluation tasks. Proposed upside: Improve teacher/tool behavior before paying for more generated data.

Failure and preservation risks. Prompt overfitting, evaluator leakage, long prompts and hidden trace/teacher costs. Forgetting: Not applicable to frozen weights; changed prompt can still regress behavior. Instability: Prompt-search instability rather than gradient instability.

Implementation and data. Complexity: Medium. Needs: Experience traces with errors, outcomes and costs. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile E). Compute: Teacher/agent calls, tool execution, judging and search dominate; frozen policy alone does not make the workflow cheap. VRAM: Local solver fit is separate; external solver memory is not part of the 8 GiB host but its monetary cost must be counted. Host RAM: Sandbox/container processes and artifacts may dominate. Storage: Environment images and outputs are additional to dataset parquet; source descriptors do not bound total storage.

Test and interaction. Fixed base model, same tools, original versus learned prompt on fresh tasks. Upstream of SFT; overlaps Dream-RSI at control level, not LoRA.

Falsifier / stop signal. No new-task improvement or gain disappears after including prompt tokens and trace collection cost.

B25. SWA bounded Replay

Track: Project KV-cache recomputation. **Priority:** LATER / systems. **Source boundary:** HISTORY.

Mechanism and evidence. Here SWA means sliding-window attention. Retain selected global/sink cache state and reconstruct omitted recent local state by replaying a bounded tail. It is not stochastic weight averaging or rehearsal training. Historical project mechanism retrieved from prior context; the separate repository branch was not freshly audited in this report.

Assumptions and expected benefit. A short suffix reproduces sufficiently useful local state; deeper receptive fields and global dependencies prevent blanket exactness claims. Proposed upside: Trade persistent cache bytes for recomputation.

Failure and preservation risks. Replayed states differ from originals; hidden boundary dependencies or duplicate writes corrupt generation. Forgetting: No parameter training; approximate state can still cause capability regression. Instability: Not an optimizer intervention.

Implementation and data. Complexity: High. Needs: Long sequences and exact-cache references for parity/quality measurement. Repository boundary: Historical RineyaArchitecture framework; not current ChatBubble native evidence.

Resources (profile F). Compute: Inference engineering: measure all predictor/draft/replay work and concurrency. VRAM: Intended cache/work savings are host- and workload-dependent; cannot infer them from paper percentages. Host RAM: Runtime buffers, retained caches and host offload need explicit ceilings. Storage: Profiling traces and separate runtime versions; no new full checkpoint inherently required.

Test and interaction. Peak/persistent bytes, recompute time, token drift and retrieval/correctness; exact and approximate modes separated. Different memory lever from PagedAttention; architecture-aware combination only.

Falsifier / stop signal. Savings vanish after replay cost or output degradation exceeds the registered margin.

B26. Never Give Up

Track: Adaptive RL sampling. **Priority:** NEXT. **Sources:** [P23,R8].

Mechanism and evidence. Allocate more attempts to currently unsolved tasks; published asynchronous sampling continues until success. Local adaptation uses bounded retries, staleness handling and explicit spending. Published LLM-RL work, distinct from the 2020 NGU namesake; local factorial is a logical-async CPU fixture.

Assumptions and expected benefit. Hard tasks remain solvable; verifier is reliable; missing successes must not trigger unbounded retries. Proposed upside: Avoid spending nearly all learning on already-easy tasks.

Failure and preservation risks. Rare-success tasks monopolize generation; retry stopping biases data if estimator is inconsistent. Forgetting: Medium to high without replay; emphasis can drift domains. Instability: High under asynchronous lag and sparse positives.

Implementation and data. Complexity: High. Needs: Verifiable tasks with complete failed/successful attempts and behavior-policy identities. Repository boundary: RNGF-001 CPU logical-async adaptation, not a native concurrent trainer.

Resources (profile B). Compute: Generation + policy scoring/backward + reference scoring + verification + evaluation; often far more than SFT. VRAM: Model, activations, logits and rollout KV; serialize trajectories on 8 GiB. CPU reference exchanges VRAM for RAM/time. Host RAM: Frozen CPU reference and trajectory buffers can be binding; bounded queues required. Storage: Store raw token IDs, behavior logprobs, rewards, policy versions, checkpoints and spending ledgers.

Test and interaction. Equal generated-token budget, task coverage, tail gains and staleness; failed attempts included. Alternative/interaction to Reinforce-Ada allocation; test each with one fixed objective first.

Falsifier / stop signal. No harder-task benefit at fixed compute or a few impossible tasks dominate spending.

B27. PagedAttention

Track: KV memory management. **Priority:** NEXT / after profiling. **Sources:** [P24].

Mechanism and evidence. Manage KV cache in blocks with noncontiguous allocation and sharing to reduce waste during serving. Established serving systems approach; not parameter pruning or an RL learning rule.

Assumptions and expected benefit. Current bottleneck is cache fragmentation or batching; backend supports the exact architecture and sampler. Proposed upside: More concurrent rollouts or lower memory waste.

Failure and preservation risks. Integration overhead, unsupported custom models and changed numerical/logprob behavior. Forgetting: No intended weight change; numerical differences must be measured. Instability: No optimizer change, but rollout/trainer mismatch can destabilize RL.

Implementation and data. Complexity: Medium to high. Needs: Representative real sequence lengths and concurrency. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile F). Compute: Inference engineering: measure all predictor/draft/replay work and concurrency. VRAM: Intended cache/work savings are host- and workload-dependent; cannot infer them from paper percentages. Host RAM: Runtime buffers, retained caches and host offload need explicit ceilings. Storage: Profiling traces and separate runtime versions; no new full checkpoint inherently required.

Test and interaction. Wall time, cache utilization, sample distribution and teacher-forced likelihood parity. Can support GRPO/NGU throughput; not interchangeable with paged optimizer states.

Falsifier / stop signal. No measured throughput/capacity benefit or likelihood mismatch breaks qualification.

B28. Score-centering off-policy RL

Track: Numerical/off-policy stability. **Priority:** NEXT / conditional. **Sources:** [P25].

Mechanism and evidence. Correct score drift associated with training–inference mismatch; it is not ordinary within-group reward or advantage centering. Published stabilization method; value depends on the mismatch actually present in the run.

Assumptions and expected benefit. Observed behavior/trainer discrepancy matches the paper setting; raw scores are retained. Proposed upside: Stabilize quantized, stale or split-backend RL after a valid synchronous baseline exists.

Failure and preservation risks. Mask a broken tokenizer or sampler contract by treating a semantic bug as small numeric drift. Forgetting: Does not directly preserve prior tasks. Instability: Intended reduction in instability; added correction can be harmful if misapplied.

Implementation and data. Complexity: Medium to high. Needs: Matched rollout/training likelihood records and known controlled mismatch conditions. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile B). Compute: Generation + policy scoring/backward + reference scoring + verification + evaluation; often far more than SFT. VRAM: Model, activations, logits and rollout KV; serialize trajectories on 8 GiB. CPU reference exchanges VRAM for RAM/time. Host RAM: Frozen CPU reference and trajectory buffers can be binding; bounded queues required. Storage: Store raw token IDs, behavior logprobs, rewards, policy versions, checkpoints and spending ledgers.

Test and interaction. First establish mismatch diagnostics; then correction ON/OFF with identical trajectories. Potentially complements importance sampling; not a substitute for correct probability accounting.

Falsifier / stop signal. No stability/quality advantage under reproduced mismatch or correction hides invalid trajectories.

B29. GPO for RL scaling

Track: Unresolved method identity. **Priority:** BLOCKED / IDENTIFY. **Source boundary:** UNRESOLVED.

Mechanism and evidence. The acronym is insufficient to identify one intended algorithm. This report does not replace it with GRPO, GSPO, a generalized preference objective or ScaleRL. Targeted web and history retrieval did not establish the intended title, equation or identifier.

Assumptions and expected benefit. A canonical primary reference and exact objective must be supplied or recovered before assessment. Proposed upside: Unknown until identified.

Failure and preservation risks. Implementing a plausible namesake and claiming the intended experiment was tested. Forgetting: Unknown. Instability: Unknown.

Implementation and data. Complexity: Unknown. Needs: Unknown; depends on objective. Repository boundary: No reliable current implementation mapping established.

Resources (profile U). Compute: Unknown; no run authorized. VRAM: Unknown. Host RAM: Unknown. Storage: Unknown.

Test and interaction. Identity gate: title, authors/version, objective and reference code before any run. Do not combine an unidentified method with a validated baseline.

Falsifier / stop signal. No efficacy hypothesis is registered until identity is resolved.

B30. Reinforce-Ada

Track: Adaptive sampling for RL. **Priority:** NEXT. **Sources:** [P26].

Mechanism and evidence. Adapt rollout allocation and form useful reward-diverse groups rather than blindly using the same sampling effort everywhere. Published adaptive sampling framework; algorithmic advantage depends on full cost accounting.

Assumptions and expected benefit. Difficulty estimates are informative, grouping semantics valid and global statistics not leaked from evaluation. Proposed upside: More learning signal per generated token.

Failure and preservation risks. Select only favorable groups, undercount discarded samples, distort the baseline or starve task families. Forgetting: Medium unless separate preservation training remains. Instability: Medium to high from adaptive estimation and nonstationarity.

Implementation and data. Complexity: Medium to high. Needs: Raw per-prompt rollout outcomes and fixed verifier. Repository boundary: No native Rineya efficacy established by this audit.

Resources (profile B). Compute: Generation + policy scoring/backward + reference scoring + verification + evaluation; often far more than SFT. VRAM: Model, activations, logits and rollout KV; serialize trajectories on 8 GiB. CPU reference exchanges VRAM for RAM/time. Host RAM: Frozen CPU reference and trajectory buffers can be binding; bounded queues required. Storage: Store raw token IDs, behavior logprobs, rewards, policy versions, checkpoints and spending ledgers.

Test and interaction. Compare fixed-group sampling at same generation budget and same policy loss; report coverage and all costs. Related to NGU, but not automatically complementary; isolate sampling first.

Falsifier / stop signal. Signal-per-token gains do not produce held-out gains or depend on excluded costs.

B31. Native Sparse Attention

Track: Trainable sparse attention. **Priority:** LATER. **Sources:** [P27].

Mechanism and evidence. Combine compressed, selected and local-window attention paths using hardware-aligned sparse computation. Published natively trainable long-context design; replacing a pretrained dense attention block is a separate conversion problem.

Assumptions and expected benefit. Kernels, layout and training adaptation fit the exact small model. Sparse selections preserve relevant evidence. Proposed upside: Potential long-context efficiency and lower attention work.

Failure and preservation risks. Conversion loss, missing retrieval paths, extra kernel complexity and little benefit at short contexts. Forgetting: High for an unqualified retrofit. Instability: Medium to high due to changed attention/selection dynamics.

Implementation and data. Complexity: High. Needs: Long-context adaptation plus adversarial evidence-position tests. Repository boundary: NSA branch enumerated only.

Resources (profile C). Compute: Additional model visits, objective changes or conversion training; no defensible fixed multiplier without prototype profiling. VRAM: Recurrent activations/new states can dominate. Checkpointing changes time/memory; 24–48 GiB may be operationally easier, not guaranteed necessary. Host RAM: Model/teacher state and preprocessing can exceed constrained host RAM. Storage: Separate architecture checkpoints and exact runtime/config artifacts; do not overwrite the base.

Test and interaction. Quality versus visible context, wall time, memory and dense-control fairness. Competing alternative to MoBA; keep out of initial short-SFT pilot.

Falsifier / stop signal. No better native quality/compute frontier or poor transfer after conversion.

B32. LoopSpec

Track: Looped-model speculative decoding. **Priority:** LATER / dependency-gated. **Sources:** [P28].

Mechanism and evidence. Use early loop states to propose tokens and pipeline verification by the complete looped target; exactness depends on the published acceptance/verification procedure. Published self-speculative decoding method for looped architectures; it does not create a competent recurrent model.

Assumptions and expected benefit. Rineya target already supports loop-depth exits, consistent caches and exact target scoring. Proposed upside: Lower latency for an already-useful looped model.

Failure and preservation risks. Low draft acceptance, bookkeeping overhead, incorrect sampling acceptance or target cache contamination. Forgetting: No intended weight update; output distribution equivalence must be demonstrated. Instability: Not a training optimizer.

Implementation and data. Complexity: High. Needs: Target-model traces at representative lengths and loop counts. Repository boundary: LoopSpec branch enumerated only.

Resources (profile F). Compute: Inference engineering: measure all predictor/draft/replay work and concurrency. VRAM: Intended cache/work savings are host- and workload-dependent; cannot infer them from paper percentages. Host RAM: Runtime buffers, retained caches and host offload need explicit ceilings. Storage: Profiling traces and separate runtime versions; no new full checkpoint inherently required.

Test and interaction. Exact greedy output equality, statistical sampling checks, acceptance and end-to-end speed; include draft cost. Depends on recurrence; separate from recurrence efficacy and attention redesign.

Falsifier / stop signal. No net speedup or failure of target-distribution/greedy equivalence.

Appendix C. Data statistics register and local profiling handoff

Quantity	Status	Value / next action
Release identity	PROVIDER METADATA	Pin the observed commit in Section 2; verify it before enrollment.
Total task rows	PROVIDER METADATA	Displayed count in executive summary; not a local independent census.
Files and asset storage	PROVIDER METADATA	Rounded tree sizes, not token volume or Docker disk forecast.
Exact per-family counts	UNMEASURED	Run a complete allowed-file census with explicit scope.
Prompt lengths / tokens	UNMEASURED	Profile with exact local tokenizer; final chat-template counts separately.
Teacher response / reasoning lengths	NOT ESTABLISHED	Need actual generated/available assistant trajectories first.
Number of attempts and success rates	UNMEASURED	Collect controlled independent rollouts with raw receipts.
Parent and benchmark overlap	UNMEASURED	Complete ancestry, source-family audit, normalized and semantic checks.
Difficulty and language proportions	UNMEASURED	Define label/proxy, sample design and uncertainty before reporting.
Complete code.parquet integrity	NOT LOCALLY VERIFIED	Provider SHA recorded in [M3]; binary fetch failed.

Example invocation on a separately acquired and authorized local subset:

```
python supporting_materials/profile_dataset.py local_subset.jsonl \
  --max-rows 10000 --output new_profile.json
```

A local tokenizer can be supplied with `--tokenizer-dir`. The profiler uses no model weights, downloads and remote code. It counts an explicitly defined untemplated role/content rendering; native chat-template qualification is a distinct step. Parquet support requires a preinstalled pyarrow library and was not exercised here. Four JSONL/normalization/budget mechanics tests passed on synthetic data; their receipt is included. No synthetic statistic is presented as MiMo measurement.

Appendix D. Candidate configurations and package inventory

`Rineya_MiMo_Experiment_Matrix.csv` contains twelve candidate pipelines with comparison, data, tokens, scope, optimizer, cost assumptions, confidence, effort, risk, information value and success/failure gates. `supporting_materials/Rineya_Method_Assessment_Matrix.csv` expands all 32 methods. Neither file authorizes execution.

`minimum_experiment_proposal.json` preserves E1's proposed configuration with unresolved checkpoint/runtime fields set to null and authorization false. It is intentionally not passed off as a compatible PRLC configuration. `references.json` stores primary publication and immutable repository anchors. `profile_dataset.py`, its synthetic tests and `profiling_validation.json` support reproducible characterization on later authorized local inputs. The original user research brief is also retained as supplied, distinct from the authored report.

References

All web/connector sources were consulted on 27 September 2026. References provide a source for a claim, not evidence that this report reproduced the method. Private repository URLs require the user's GitHub access. Mutable release pages should be replaced by frozen revision/file hashes in an executable experiment manifest.

[M1] XiaomiMiMo, MiMo-V2.6-RL-oss dataset card and displayed viewer metadata. Official release; inspected 2026-09-27. Viewer statistics are provider-reported, not a local census.

https://huggingface.co/datasets/XiaomiMiMo/MiMo-V2.6-RL-oss

[M2] MiMo release tree and commit. Observed main revision; individual pages can have independent cache dates. Freeze and recheck this revision before use.

https://huggingface.co/datasets/XiaomiMiMo/MiMo-V2.6-RL-oss/commit/639865fd3374018d6cb29b9fb82dd531406fcf5f

[M3] MiMo code.parquet metadata. Displayed SHA-256 e15733cf2451cfbc5492a4120f7f8cfddbad818aa9f0b324c79888dd1fece161; binary not downloaded.

https://huggingface.co/datasets/XiaomiMiMo/MiMo-V2.6-RL-oss/blob/main/code.parquet

[M4] MiMo release tree: general assets. General contains train.parquet and envs; not a teacher-answer token census.

https://huggingface.co/datasets/XiaomiMiMo/MiMo-V2.6-RL-oss/tree/main/general

[P01] MiniCPM: Unveiling the Potential of Small Language Models with Scalable Training Strategies. 2024; WSD reference.

https://arxiv.org/abs/2404.06395

[P02] LoRA: Low-Rank Adaptation of Large Language Models. 2021; foundational parameterization.

https://arxiv.org/abs/2106.09685

[P03] Maximum Likelihood Reinforcement Learning. 2026; MaxRL. Paper/author project consulted; practical estimator must be version-bound.

https://arxiv.org/abs/2602.02710

[P04] Rotary-Positioned Low-Rank Linear Attention for Efficient Diffusion Transformers. 2026; RoLA, video diffusion attention.

https://arxiv.org/abs/2609.06712

[P05] Direct Preference Optimization: Your Language Model is Secretly a Reward Model. 2023; pairwise preference objective.

https://arxiv.org/abs/2305.18290

[P06] Scalable Class-Incremental Continual Learning via a Sparsity-Based Framework. 2026; intended sparsity-based CLARE. Search-indexed primary abstract; direct page fetch failed.

https://arxiv.org/abs/2609.17026

[P07] Recursive Self-Improvement through Evolving Worlds. 2026; Dream-RSI, exploration orchestration.

https://arxiv.org/abs/2609.14858

[P08] Sparse Adapters for Continual Learning in Large Language Models. 2026; JumpLoRA.

https://arxiv.org/abs/2604.16171

[P09] Scaling Distribution-Matching RL to Large Language Models. 2026; GFlowRL.

https://arxiv.org/abs/2607.13394

[P10] Deja Vu: Contextual Sparsity for Efficient LLMs at Inference Time. 2023; input-dependent inference sparsity, not recurrence.

https://arxiv.org/abs/2310.17157

[P11] F-WANDA: Fisher-Reweighted Post-Training Pruning for Sustainable Deployment of Large Language Models. 2026; row-budget allocation, not a proved delta-retention algorithm.

https://arxiv.org/abs/2608.00481

[P12] Soft Tokens, Hard Truths. 2025; continuous-CoT RL with embedding noise.

https://arxiv.org/abs/2509.19170

[P13] Thinking with Looped Flows. 2026; learned local denoising and recurrent latent refinement.

https://arxiv.org/abs/2609.11801

[P14] DeepLoop: Depth Scaling for Looped Transformers. 2026; tied-depth Post-LN residual scaling. v2 inspected.

https://arxiv.org/abs/2607.13491

[P15] Large Language Diffusion Models. 2025; LLaDA, representative discrete diffusion language modeling.

https://arxiv.org/abs/2502.09992

[P16] Full-Bandwidth Transformer. 2026; recurrent hidden-state feedback across tokens.

https://arxiv.org/abs/2608.08888

[P17] Explorative Modeling: Unlocking a Third Pretraining Axis and End-to-End Generation. 2026; candidate exploration in training.

https://arxiv.org/abs/2607.27372

[P18] Equilibrium Reasoners: Learning Attractors Enables Scalable Reasoning. 2026; task-conditioned attractor learning.

https://arxiv.org/abs/2605.21488

[P19] Nexus: Same Pretraining Loss, Better Downstream Generalization via Common Minima. 2026; common-minima optimizer, v2 abstract inspected.

https://arxiv.org/abs/2604.09258

[P20] ReAct: Synergizing Reasoning and Acting in Language Models. 2022; interleaved reasoning, actions and observations.

https://arxiv.org/abs/2210.03629

[P21] MoBA: Mixture of Block Attention for Long-Context LLMs. 2025; block-routed attention.

https://arxiv.org/abs/2502.13189

[P22] Combee: Scaling Prompt Learning for Self-Improving Language Model Agents. 2026; prompt learning, not weight optimization.

https://arxiv.org/abs/2604.04247

[P23] Learning to Solve Hard Problems in RL for LLMs by Never Giving Up. 2026; NGU adaptive sampling. Distinct from the earlier identically named exploration algorithm.

https://arxiv.org/abs/2609.13443

[P24] Efficient Memory Management for Large Language Model Serving with PagedAttention. 2023; KV-memory management and serving.

https://arxiv.org/abs/2309.06180

[P25] Score Centering Stabilizes Off-policy Reinforcement Learning. 2026; score correction for training/inference mismatch.

https://arxiv.org/abs/2609.20807

[P26] Reinforce-Ada: An Adaptive Sampling Framework for Reinforce-Style LLM Training. 2025; adaptive rollout allocation.

https://arxiv.org/abs/2510.04996

[P27] Native Sparse Attention: Hardware-Aligned and Natively Trainable Sparse Attention. 2025; NSA.

https://arxiv.org/abs/2502.11089

[P28] LoopSpec: Pipelined Self-Speculative Decoding for Looped Transformers. 2026; primary indexed abstract consulted; direct fetch failed.

https://arxiv.org/abs/2609.17184

[P29] QLoRA: Efficient Finetuning of Quantized LLMs. 2023; memory-efficient adaptation.

https://arxiv.org/abs/2305.14314

[P30] DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. 2024; GRPO reference.

https://arxiv.org/abs/2402.03300

[P31] DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. 2025; reinforcement learning and distilled students, not evidence for a 344M reproduction.

https://arxiv.org/abs/2501.12948

[P32] LIMA: Less Is More for Alignment. 2023; selective SFT evidence, not a universal small-data scaling law.

https://arxiv.org/abs/2305.11206

[P33] The Art of Scaling Reinforcement Learning Compute for LLMs. 2025; ScaleRL; not established as the requested GPO identity.

https://arxiv.org/abs/2510.13786

[P34] Can Large Reasoning Models Self-Train?. 2025; self-reward improvement and eventual reward-hacking collapse.

https://arxiv.org/abs/2505.21444

[P35] Continual Learning for Vision-Language-Action Models via Autonomous Adapter Routing and Expansion. 2026; different CLARE namesake, not the intended class-incremental sparsity paper.

https://arxiv.org/abs/2601.09512

[R1] PRLC-001 README. Private repository; retrieved at immutable commit, 2026-09-27. Statements about tests are recorded results, not tests rerun for this report.

https://github.com/LUNCY579/ChatBubble/blob/5517361ec09e02308eae0352b2e53a70853c5245/experiments/progressive_rl_curriculum_v1/README.md

[R2] PRLC-001 validation record. Private repository; retrieved at immutable commit, 2026-09-27. Statements about tests are recorded results, not tests rerun for this report.

https://github.com/LUNCY579/ChatBubble/blob/5517361ec09e02308eae0352b2e53a70853c5245/experiments/progressive_rl_curriculum_v1/VALIDATION.json

[R3] PRLC-001 objective source. Private repository; retrieved at immutable commit, 2026-09-27. Statements about tests are recorded results, not tests rerun for this report.

https://github.com/LUNCY579/ChatBubble/blob/5517361ec09e02308eae0352b2e53a70853c5245/experiments/progressive_rl_curriculum_v1/objectives.py

[R4] PRLC-001 integration reconnaissance. Private repository; retrieved at immutable commit, 2026-09-27. Statements about tests are recorded results, not tests rerun for this report.

https://github.com/LUNCY579/ChatBubble/blob/5517361ec09e02308eae0352b2e53a70853c5245/experiments/progressive_rl_curriculum_v1/RECONNAISSANCE.md

[R5] CSS-001 README. Private repository; retrieved at immutable commit, 2026-09-27. Statements about tests are recorded results, not tests rerun for this report.

https://github.com/LUNCY579/ChatBubble/blob/268eb97520f2165280287425ddb8ca66e01ed49f/experiments/rineya_composable_subspace_stack_v1/README.md

[R6] SOUP-MINI-001 README. Private repository; retrieved at immutable commit, 2026-09-27. Statements about tests are recorded results, not tests rerun for this report.

https://github.com/LUNCY579/ChatBubble/blob/816a1e8f4d1ab1c9d1ab847697d962a141521bac/experiments/titan_model_soup_mini_v1/README.md

[R7] MAXRL-COMP-002 Q0 qualification. Private repository; retrieved at immutable commit, 2026-09-27. Statements about tests are recorded results, not tests rerun for this report.

https://github.com/LUNCY579/ChatBubble/blob/21532b765269b19e6aabf8d596e7409be51b37ea/experiments/maxrl_comp002_pretraining_v1/QUALIFICATION.md

[R8] RoLA/NGU factorial README. Private repository; retrieved at immutable commit, 2026-09-27. Statements about tests are recorded results, not tests rerun for this report.

https://github.com/LUNCY579/ChatBubble/blob/a99f3bc4195526289ef6daedb18b9c9bed38bbe5/experiments/rola_ngu_factorial_v1/README.md

[R0] ChatBubble branch inventory. 35 refs returned on 2026-09-27. Default main head b0ca54bdc1b69e2b19d7baeca2e3fdde3ff0a7b2. Branch existence is not qualification.

https://api.github.com/repos/LUNCY579/ChatBubble/branches?per_page=100&page=1

If I were running the Rineya project next

Experiment 0 — qualify the source and the measurement. Freeze the exact student, local tokenizer, runtime, permitted training parents and deterministic verifier. Run zero-update native parity, then a separately authorized bounded backward/resume fixture. **Stop** on identity, masking, likelihood, verifier or resource failures; do not lower gates to obtain a run.

Experiment 1 — establish short-task acquisition. Run the fixed-scope E1 LoRA SFT proposal against the unchanged base, first as an operational screen and then as a three-seed pilot only when justified. **Stop scaling** if capability does not generalize or preservation fails. A loss curve alone is not a pass.

Decision gate — diagnose rather than stack methods. Acquisition plus preservation leads to E2A, an order-only curriculum comparison. Acquisition with forgetting leads to E2B, an isolated preservation comparison. Underfitting leads to E2C, a data/runtime check followed only then by a scope comparison. Memorization leads to new task families, not repeated epochs on exposed examples.

Experiment 3 — choose the learning objective. Once verified on-policy successes exist, compare rejection-SFT, GRPO and MaxRL from the same SFT parent using equal total rollout budgets and one fixed sampler. **Stop** any arm with broken behavior probabilities, insufficient useful signal, uncounted failures or unacceptable retention losses. Test NGU or Reinforce-Ada only as a subsequent sampling factor.

Experiment 4 — test modularity or actual MiMo transfer. For modularity, compare a mixed LoRA with separately acquired skills, dose-matched composition controls and input-only routing. For MiMo transfer, qualify a small isolated code environment and fresh agent tasks before collecting expensive trajectories. **Stop** when an ineffective donor, incompatible architecture or unqualified grader makes the comparison uninterpretable.

Full-training candidate — only a frozen, confirmed winner. Expand data and compute only after an independently evaluated recipe improves capability while meeting a prospectively specified preservation margin. Keep the final bank unopened until that candidate is frozen. The result should determine the next expenditure of compute—not merely add another implemented but unqualified branch.